1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the high-level functional and structural requirements of the **Whizard Talent-Building Ecosystem**.

Enterprises today face a structural talent problem: **they cannot reliably convert learning spend into workforce readiness or hiring confidence**. Traditional universities, generic courses, static LMS platforms, and interview-driven hiring all fail to produce predictable, job-ready talent.

Whizard is conceived as a **AI-native workforce readiness platform** that replaces traditional L&D, reduces hiring risk, and builds a continuous talent pipeline from colleges to enterprises using a **Workforce Readiness Capability Framework (WRCF)**. Unlike traditional courses, Whizard uses multi-agent AI and work-object simulations to generate verifiable evidence of mastery.

Whizard is a multi-application digital platform designed to build structured, measurable workforce readiness within colleges through long-term, project-first mastery clubs. The system integrates students, colleges, industry clubs, and companies into a unified capability-driven ecosystem.

This document defines the scope, actors, system boundaries, operating guardrails, and high-level interdependencies between the applications that constitute Whizard.

This SRS serves as the reference document for:

- Product Owners
- System Architects
- Development Teams
- UI/UX Designers
- Testing Teams
- Implementation and Deployment Teams

1.2 Scope

Whizard is a structured, role-based ecosystem that operationalizes student development across four progressive stages:

- Exploration
- Skill Foundation

- Mastery
- Industry Integration

The system consists of four integrated but role-segregated applications:

1. Club Admin App
2. Student App
3. College App
4. Company App

The core objective of Whizard is to:

- Build early performance signals for students before graduation
- Standardize industry-aligned capability frameworks
- Enable structured skill progression tracking
- Facilitate internship and employment pipelines
- Reduce hiring friction and onboarding time
- Create a sustainable campus talent pipeline

Whizard operates on a Work-Object-Centric Workforce Readiness Capability Framework (WRCF) structured as:

Sector
Industry
→ Functional Group
→ Primary Work Object (PWO)
→ Secondary Work Object (SWO)
→ Capability
→ Proficiency
→ Skill
→ Task
→ Control Point
→ Evidence
→ Evaluator

As per WRCF

- Capability Instances belong to the Industry.
- Competence belongs to the Role.
- People are assessed against Role Competence.
- Role competence is achieved through verified Capability Instances.

- A Capability Instance = SWO + Capability + Proficiency.

WRCF standardizes:

- Industry definitions
- Role definitions
- Functional groups
- Capability structures
- Skill evidences
- Certification pathways

Whizard is not a job portal. It is a capability-governed talent incubation and industry-integration platform.

1.3 Definitions, Acronyms, and Abbreviations

Whizard – A structured college-based talent development ecosystem.

Parent Club – An industry-domain-level governance entity (e.g., Manufacturing, Finance, AI).

Club – A structured learning and mastery unit under a Parent Club. Typically, not necessarily, operates at the Industry Function group.

Certification Level – Progressive levels within a club reflecting increasing competence and mastery.

Exploration Stage – Stage where students can explore up to three clubs.

Mastery Stage – Stage where students focus on one primary club and complete advanced certifications and projects.

Industry Integration Stage – Stage where students engage with companies through internships, projects, and conditional training programs.

Workforce Readiness Capability Framework — A structured model defining roles, capabilities, skills, control points, and evidence aligned to industry standards.

Sector: A broad economic grouping of industries; provides macro-level context; not an organization or curriculum.

1️⃣ Industry

Definition:
A broad economic domain in which organizations operate, defined by a common production logic, regulatory structure, and value chain.

Purpose in WRCF:
Provides the **contextual ecosystem** within which capability models are built.

Examples:

Thermal Power Generation

Retail Banking

IT Software Development

Key Principle:
Industry defines *what kind of work exists*, not who performs it.

2 Company

Definition:
A specific legal entity operating within an industry that has its own asset base, processes, standards, and operating philosophy.

Purpose in WRCF:
Company is where **industry framework gets customized** (SOPs, equipment make, compliance rules).

Key Distinction:

Industry = generic capability model

Company = contextual implementation of that model

Example:

Industry: Thermal Power

Company: NTPC / Tata Power / State GENCO

3 Functional Group

Definition:
A major operational or support domain within an industry where related systems and processes are grouped.

Purpose in WRCF:
Organizes work into manageable operational clusters.

Examples (Thermal Power):

Coal Handling Plant (CHP)

Boiler

Turbine

Generator

Electrical Systems

C&I

Ash Handling

Key Insight:
Functional Group = where work happens.

4 Primary Work Object (PWO)

Definition:
A major system, process cluster, or operational domain within a functional group that delivers a distinct operational purpose.

Purpose in WRCF:
Defines the **competency boundary** for capability modeling.

Examples (CHP):

Coal Receiving System

Conveying Systems

Crushing & Sizing

Stacking & Reclaiming

Dust, Environment & Safety

Key Insight:
PWO = What major system must be mastered.

5 Secondary Work Object (SWO)

Definition:
A specific physical system, equipment, or sub-process within a Primary Work Object.

Purpose in WRCF:
Enables granular capability and task definition.

Examples (Crushing & Sizing PWO):

Coal Crusher

Crusher Drive System

Crusher Rotor

Vibrating Screen

Screen Drive

Key Insight:
SWO = Where tasks are actually performed.

6 Capabilities

Definition:
Observable, role-agnostic abilities required to effectively interact with a work object.

Capabilities describe *how a person engages with a system*, not what equipment they touch.

Code	Capability
CAP-01	Work Object Familiarization
CAP-02	Process Familiarization (Input → Output)
CAP-03	Operation Execution
CAP-04	Monitoring & Exception Detection
CAP-05	Fault / Error Recognition
CAP-06	First Response & Resolution

CAP-07	Preventive / Proactive Action
CAP-08	Optimization & Judgment
CAP-09	Compliance & Safety Awareness
CAP-10	Documentation, Communication & Handover

Key Insight:
Capability = Type of engagement with system.

Capability vs Competence

Capability is a reusable human action ability that can be trained and developed independent of role or context while

Competence is the verified ability to apply one or more capabilities effectively, safely, and consistently in a specific role, asset, and process context.

```java
Industry Capability Framework
 ├── Capability (CAP-01, CAP-02, …)
 │    └── Proficiency Levels (L1–L4)
 │         └── Skill Units (atomic tasks)
 │
 └── Role Definition
      └── Required COMPETENCE
           ├── Asset / Process context
           ├── Required Capabilities
           ├── Required Proficiency
           └── Evidence rules
```

So:

- **Capabilities belong to the industry**
- **Competence belongs to the role**
- **People are assessed against competence**

Proficiency

Definition

The degree of autonomy, judgment, and responsibility with which a capability is exercised.

Proficiency Level	Label	Description	Risk Exposure
L1	Plant Awareness	Knowledge only	Zero
L2	Assisted Execution	Perform under observation	Zero
L3A	Conditional Independence – Supervised	Perform alone, supervisor nearby	Low
L3B	Conditional Independence – Scoped	Perform within boundaries	Controlled

L4	Full Independence	Asset/ Service -level autonomy	Institutionally backed

L1 – Awareness - Validate conceptual understanding before physical execution

System shall assign plant process learning modules linked with respective role in relation to requisite:

Functional Group
Asset / Service
Capability

The system shall conduct knowledge assessment before L2 unlock.

L2 – Assisted Execution

Demonstrate controlled task execution under supervision. ("Show me you can do it")

During Internship creation, based on the role attached to Internship, User can select predefined Skills and respective tasks while creating Weekly Schedule

Student shall log execution attempt and capture respective evidence against each task. Mentor shall validate performance live or post-task. System shall require X successful supervised attempts before successful completion of task.

Risk:
Supervisor exposure = zero (observational only).

L3A Conditional Independence

Test low-risk autonomy with standby supervision. ("Do it, I'm nearby")

Typically in the initial weeks of Internships Only low-risk tasks are being assigned

Mentor should remain "On Standby" during execution. System shall enforce minimum successful repetition threshold. Safety violation shall auto-reset stage progression.

Risk:
Supervisor risk = low and controlled.

L3B– Conditional Independence (Scoped)

Independence is boundary-constrained. ("You can do this — within boundaries")

System shall define operational boundaries per asset:

Example parameters:

Temperature range
Vibration limit
Misalignment tolerance
Pressure threshold

Student shall log independent task performance. The system shall define escalation triggers. System shall record:

Incident-free execution
Correct escalation events
Time-to-escalation
Decision quality

Risk:
Shared between supervisor and system.

Key rule

Proficiency is **how well**, not **what.** It is applied to **capabilities**. It is used to **define trust boundaries**

📌 **Proficiency answers: "How independently can this action be performed?"**

7 Skill

Definition:
A specific, demonstrable ability that enables a person to perform a defined action under a given capability for a given work object.

Structure:
Skill =
(PWO + SWO + Capability + Proficiency)

Example:

"Start crusher drive motor as per SOP"
"Identify belt misalignment condition"
"Monitor dust extraction airflow trend"

Important Clarification:
Skills are operationally meaningful.
Skills are validated through tasks.

```
PWO
 └── SWO
      └── Capability (CAP-01 … CAP-10)
           └── Proficiency (L1–L4)
                └── Skill (Capability-level)
                     └── Task (Internship-executable)
                          └── Evidence
```

🔎 How Everything Connects (Hierarchy View)

```
Industry
   → Functional Group
      → Primary Work Object (PWO)
            → Secondary Work Object (SWO)
            → Capability
              → Proficiency
                  → Skill
                       → Task
                         → Control Point
                                 → Evidence
                               → Evaluator
```

Task

📌 Definition

A **Task** is a clearly defined, observable unit of work performed on a specific Secondary Work Object to demonstrate a specific Skill under a defined Capability and Proficiency level.

It is the **smallest executable field action** that can be:

Assigned during internship
Performed on the shop floor
Observed by a supervisor
Logged in the system
Validated objectively

Structure of a Task

Task = (PWO + SWO + Skill + Context + Mode of Execution)

Example:

“Start crusher drive motor as per SOP and record start-up current”

Control Points

Definition:
Observable validation checkpoints that confirm whether a skill has been correctly demonstrated.

Control Points define:

What must be verified
What evidence is acceptable
Who validates it

Examples:

SOP followed step-by-step
Alarm acknowledged within time limit
Inspection recorded in logbook
Hazard escalated as per protocol

Control Point Structure:
Skill → Task → Control Point → Evidence → Evaluator

Key Insight:
Control Points create trust.

They convert "He knows" into
"He has demonstrated and been validated."

Tasks build evidence

Evidence validates skills

Validated skills authorize work

Evidence

📌 Definition

Evidence is the documented or observable proof that a Task was executed correctly and that the associated Skill has been demonstrated to the required proficiency level.

Evidence converts:

"I did it" into "It is verifiably done."

📌 Evidence Types in WRCF

Evidence Type	Description	Example
Observation	Direct visual confirmation	Supervisor observed SOP compliance
Log Entry	Entry in official register	Start-up current logged
Checklist	Signed inspection checklist	Drain inspection record
Alarm Log	Control room record	Alarm acknowledged in DCS
Photo / Video	Visual record	PPE compliance photo
Drill Record	Simulation participation	Fire drill sign-off
Trend Review	Analytical validation	Vibration trend analysis
Incident Report	Structured event documentation	Root cause report
Communication Log	Coordinated action	Shift handover note

📌 Evidence Must Be:

Verifiable
Traceable
Linked to specific task
Linked to evaluator
Time-stamped
Immutable (ideally)

DESIGN PRINCIPLES (PIN THESE)

1. **Functional Groups define where work happens.**
2. **Work Objects are the anchor**
3. **Capabilities are universal**
4. **Proficiency defines autonomy**
5. **Skills are atomic actions**
6. **Control points define trust**
7. **Courses prepare; internships validate**
8. **Roles declare trust**

1.4 Overall Description

1.4.1 Product Perspective

Whizard is a distributed multi-application ecosystem operating on a unified backend data and capability architecture.

The system consists of:

- Governance Layer (Club Admin App)
- Execution Layer (Student App)
- Institutional Integration Layer (College App)
- Industry Integration Layer (Company App)

All applications are interconnected through shared data structures, user role management, and the Workforce Readiness Capability Framework.

Each application enforces strict role-based access control (RBAC) and operates within clearly defined authority boundaries.

1.4.2 Product Functions (High-Level)

The Whizard ecosystem enables:

- Creation and governance of industry-aligned clubs
- Definition and version control of capability frameworks
- Student onboarding and certification progression tracking
- Social collaboration and communication
- Course and project management
- Internship and job posting management
- Structured evaluation and feedback loops
- Institutional analytics and job readiness indexing
- Corporate training and conditional offer management
- Billing and payment management for corporate services

1.4.3 User Classes and Characteristics

The system includes the following primary user classes:

A. Club Governance Users

- System Admin
- Club Admin
- Club Corporate Manager
- Club Content Manager
- Club Communication Manager

These users define structure, standards, certifications, corporate partnerships, and communication governance.

B. Student Users

- Student
- Local Chapter Club Executive
- College / Club Alumni

Students progress through certification pathways, build project portfolios, and apply for opportunities.

C. College Users

- College Admin
- Placement Cell
- Club Coordinators

These users monitor student development, manage local chapters, and access institutional readiness analytics.

D. Company Users

- HR Head
- Recruitment Manager
- Functional Manager
- Training Manager
- Department Head
- Department Manager
- Company Admin

These users manage internships, job requisitions, training programs, evaluations, and capability customization.

1.5 Operating Environment

Whizard is intended to operate as:

- A cloud-based web and mobile-enabled multi-tenant platform
- Secure, role-based access-controlled environment
- Scalable architecture supporting multiple colleges, clubs, and companies
- Data-isolated but framework-integrated ecosystem

The system must support:

- Multi-institution onboarding
- Multi-company onboarding
- Version-controlled framework customization
- Secure document and transcript uploads
- Role-segregated analytics dashboards

1.6 Design and Implementation Constraints

The system shall adhere to the following constraints:

1. Role-Based Access Control (RBAC) must be enforced across all applications.
2. Certification advancement must require evidence-based evaluation.
3. Industry-wide framework modifications must follow version control protocols.
4. Student data visibility must be permission-driven.
5. Colleges cannot alter industry-wide capability standards.
6. Companies may customize frameworks but cannot modify industry baseline definitions.
7. Data privacy and controlled access must be maintained across all actor categories.

1.7 Assumptions and Dependencies

The Whizard system operates under the following assumptions:

- Colleges will actively onboard students and support local chapters.
- Companies will define roles aligned to the capability framework.
- Students will participate in structured progression pathways.
- Clubs will maintain updated certification structures and content pipelines.

System-level dependencies include:

- Active collaboration between Club Governance and Company stakeholders
- Accurate student participation data
- Institutional cooperation for internship evaluations
- Industry engagement in project sponsorships

The effectiveness of Whizard depends on coordinated participation across all four applications.

1.8 Document Organization

This SRS document is structured as follows:

- Section 1: Introduction
- Section 2: Overall Description
- Section 3: Specific Requirements
- Section 4: External Interface Requirements
- Section 5: Non-Functional Requirements
- Section 6: Data Model and Entity Relationships
- Section 7: Security and Compliance Requirements

2. Overall Description

2.1 Product Perspective

Whizard is a cloud-based, multi-tenant, role-driven ecosystem designed to integrate academic institutions, industry governance bodies, and companies into a structured workforce readiness pipeline.

The system is architected as four logically separated but data-integrated applications:

1. Club Admin App (Governance Layer)
2. Student App (Execution Layer)
3. College App (Institutional Integration Layer)
4. Company App (Industry Integration Layer)

All four applications operate over a centralized core platform consisting of:

- Identity & Role Management Engine
- Workforce Readiness Capability Framework (WRCF) Engine
- Certification & Progression Engine
- Project & Event Management Engine
- Internship & Job Lifecycle Engine
- Communication & Social Engine
- Billing & Subscription Engine
- Analytics & Reporting Engine

2.1.1 System Boundary

Whizard operates as:

- A centralized SaaS platform
- A shared capability governance model
- A distributed role-based application interface

External entities interacting with Whizard include:

- Colleges (Institutional Administrations)
- Companies (Industry Partners)
- Students
- Service Providers (Content & Training Teams)

The platform does not manage payroll, academic grading systems, or corporate HRMS systems directly, but may integrate with them via APIs where applicable.

2.2 Product Functions

At a high level, Whizard provides the following major functional capabilities:

2.2.1 Governance Functions (Club Admin App)

- Creation and management of Parent Clubs
- Creation and management of Clubs
- Definition and versioning of Workforce Readiness Capability Framework
- Definition of Industry-wide role structures
- Certification level design and progression logic
- Onboarding and management of Companies
- Onboarding and management of Colleges
- Onboarding and management of Content Service Providers
- Communication moderation within club ecosystems
- Corporate pricing, sponsorships, advertising configuration

2.2.2 Student Lifecycle Functions (Student App)

- Student onboarding and profile creation
- Exploration of up to three clubs
- Selection of primary club for certification pathway
- Enrollment in courses and projects
- Participation in events and contests
- Submission of project deliverables
- Capability evidence tracking
- Certification progression monitoring
- Internship and job application submission
- Participation in Whizard Social
- Mentorship interactions

2.2.3 Institutional Functions (College App)

- Student bulk onboarding
- Semester transcript uploads
- Launch and management of local club chapters
- Monitoring of student capability progression
- Job readiness index visibility
- Internship coordination and mid-term feedback interface
- Institutional analytics dashboards
- Communication with students via platform

2.2.4 Industry Functions (Company App)

- Company onboarding and profile management
- Department creation and role mapping
- Customization of Workforce Readiness Capability Framework
- Internship requisition posting
- Job requisition posting
- Project and challenge sponsorship
- Internship evaluation and scoring
- Conditional offer training program management
- Trainee assessment and evaluation
- Recruitment analytics
- Billing and payment management
- Advertisement campaign management

2.2.5 Workforce Readiness Capability Framework Model

The WRCF shall support a work-object-based hierarchy:

1. Primary Work Object (PWO) – Major industrial asset or process context.
2. Secondary Work Object (SWO) – Sub-component or operational subset.
3. Capability – Industry-defined ability.
4. Proficiency – Level of mastery (L1–L4).
5. Skill – Atomic demonstrable competency.
6. Task – Executable work action.
7. Control Point – Validation checkpoint.
8. Evidence – Proof of task execution.
9. Evaluator – Authorized supervisor validating mastery.

The system shall support Role Competence defined as:

Role
→ Assigned Capability Instances
→ Required Skills
→ Required Tasks
→ Required Evidence Rules

A person becomes “Role Ready” only when all required capability instances are verified.

2.3 User Classes and Characteristics

The Whizard system supports multiple user classes with distinct authority scopes.

2.3.1 System-Level Users

System Admin

- Full system configuration control
- Multi-tenant governance authority
- Access to global configuration and system logs

2.3.2 Club-Level Users

Club Admin

- Defines certification architecture
- Manages capability framework
- Controls industry governance structures

Club Corporate Manager

- Manages corporate partnerships
- Oversees sponsorships and pricing packages

Club Content Manager

- Manages courses, projects, training providers

Club Communication Manager

- Moderates social communication ecosystem

2.3.3 Student-Level Users

Student

- Enrolls in clubs and certifications
- Submits projects and assessments
- Applies for internships and jobs

Local Chapter Club Executive

- Limited administrative privileges within college chapter

Alumni

- Limited access for networking and mentoring

2.3.4 College-Level Users

College Admin

- Onboards students
- Manages local club chapters
- Uploads transcripts

Placement Cell

- Views student progression and job readiness data
- Coordinates with companies

Club Coordinators

- Monitor club-specific progress within institution

2.3.5 Company-Level Users

Company Admin

- Manages company page
- Configures departments and users
- Customizes framework mapping

HR Head

- Approval authority for requisitions

Recruitment Manager

- Posts internships and jobs
- Manages recruitment pipeline

Functional Manager

- Evaluates interns

Training Manager

- Manages training programs

Department Head / Department Manager

- Approves and mentors internships
- Conducts evaluations and interviews

2.4 Operating Environment

Whizard shall operate in the following environment:

- Cloud-hosted SaaS infrastructure
- Web-based application interface
- Mobile-responsive interface
- Secure multi-tenant architecture
- API-enabled integration capability

Supported environments:

- Modern web browsers (Chrome, Edge, Safari, Firefox)
- Secure HTTPS-based communication
- Role-based authentication and authorization

2.5 Design and Implementation Constraints

The system shall adhere to the following constraints:

1. Role-Based Access Control (RBAC) must be enforced across all modules.
2. Certification progression must require evidence validation.
3. Capability framework modifications must be version-controlled.
4. Multi-tenant data isolation must be maintained.
5. Personal data protection regulations must be complied with.
6. Company-level customizations must not override industry-level baseline definitions.
7. Students cannot modify evaluation outcomes.
8. Colleges cannot alter industry-defined role competence definitions.

2.6 User Documentation

The system shall provide:

- Role-specific onboarding guides
- User manuals for each application
- Video walkthroughs
- In-app guided workflows
- Help center documentation
- FAQ and support request mechanism

2.7 Assumptions and Dependencies

2.7.1 Assumptions

- Colleges will support local chapter operations.
- Companies will participate in internships and role mapping.
- Students will engage in structured progression pathways.
- Service providers will maintain updated content pipelines.

2.7.2 Dependencies

The system depends on:

- Active industry participation
- Institutional cooperation for internship evaluation
- Reliable student engagement data
- Periodic framework updates aligned to industry changes
- Secure and scalable cloud infrastructure

2.8 System-Level Guardrails

The Whizard platform operates under the following structural guardrails:

1. Governance integrity must be preserved across all modules.
2. Capability definitions are centrally structured and hierarchically controlled.
3. Evidence-based progression is mandatory.
4. No actor may access data beyond authorized scope.
5. All internship and certification outcomes must be auditable.
6. Corporate hiring access is permission-bound to application or program participation.

2.9 Strategic Alignment

Whizard is designed to:

- Reduce hiring uncertainty for companies.
- Provide measurable readiness metrics to colleges.
- Enable structured career progression for students.
- Build sustainable campus talent pipelines.

The system integrates governance, execution, institutional tracking, and industry absorption into a single capability-driven architecture.

2.10 High-Level System Architecture Overview

2.10.1 Architectural Philosophy

Whizard is designed as a **layered, capability-centric, multi-tenant SaaS architecture**.

The system separates:

- Governance
- Execution
- Institutional Monitoring
- Industry Integration

while operating over a unified core services backbone.

The architecture enforces:

- Role-Based Access Control (RBAC)
- Capability Framework Integrity
- Multi-tenant Data Isolation
- Modular Application Interfaces
- Separation of governance and execution

Whizard is:

Work-object-driven

Supervisor-validated

Evidence-backed

Task-granular

Role-competence certified

2.10.2 Logical Architecture Layers

The Whizard system consists of five primary logical layers:

1. Presentation Layer
2. Application Services Layer
3. Domain & Capability Engine Layer
4. Data & Persistence Layer
5. Integration & External Interface Layer

2.10.3 Presentation Layer (User Interface Layer)

This layer contains the four role-segregated applications:

- Club Admin App
- Student App
- College App
- Company App

Each application:

- Has role-specific dashboards
- Enforces permission-scoped UI rendering
- Interacts with shared backend services via secure APIs
- Prevents cross-role unauthorized actions

Although presented as four applications, they operate on a shared backend architecture.

2.10.4 Application Services Layer (Business Logic Layer)

This layer contains the core system engines responsible for enforcing business rules.

Major services include:

1. Identity & Access Management Service

- Authentication
- Authorization
- Role hierarchy enforcement
- Multi-tenant isolation

2. Club Governance Service

- Parent club management
- Certification architecture management
- Framework version control

3. Workforce Readiness Capability Framework Engine

- Industry structure definitions
- Functional group definitions
- Role competence mapping
- Capability & skill taxonomy
- Evidence and control point tracking

This engine acts as the central domain backbone.

4. Student Progression Engine

- Enrollment logic
- Certification advancement validation
- Evidence submission tracking
- Skill scoring

5. Internship & Job Lifecycle Engine

- Requisition creation
- Approval workflows
- Application management
- Internship evaluation
- Offer lifecycle tracking

6. Training & Assessment Engine

- Course management
- Assignment evaluation
- Conditional offer training
- Certification scoring

7. Communication & Social Engine

- Club-based feeds
- Role-scoped visibility
- Moderation workflows

8. Billing & Corporate Subscription Engine

- Package management
- Sponsorship billing
- Advertisement billing
- Invoice tracking

9. Analytics & Job Readiness Engine

- Student readiness index calculation
- Institutional dashboards
- Corporate analytics
- Certification trend analysis

2.10.5 Domain & Capability Engine Layer (Core Intellectual Layer)

This layer contains the structured Workforce Readiness Capability Framework (WRCF).

The hierarchy includes:

Industry
→ Functional Group
→ Primary Work Object (PWO)
→ Secondary Work Object (SWO)
→ Capability
→ Proficiency
→ Skill
→ Task
→ Control Point
→ Evidence
→ Evaluator

All other modules reference this hierarchy.

Key Guardrail:

The WRCF engine acts as a controlled reference model. No application can bypass or directly manipulate framework definitions without governed workflows.

2.10.6 Data & Persistence Layer

The data layer includes:

- Multi-tenant database architecture
- Role-mapped access controls
- Version-controlled framework records
- Student portfolio records
- Internship evaluation records
- Certification records
- Corporate customization records
- Audit logs

Data isolation rules:

- College data is institution-scoped
- Company data is organization-scoped
- Student data is permission-controlled
- Framework baseline definitions are centrally governed

All progression, evaluation, and approval workflows must be auditable.

2.10.7 Integration & External Interface Layer

Whizard may integrate with:

- College ERP systems (for transcripts)
- Corporate HRMS systems
- Payment gateways
- Email and notification services
- Video conferencing systems
- Assessment proctoring systems

All integrations shall occur through secured APIs with token-based authentication.

2.10.8 Actor Interaction Model (Conceptual Flow)

The high-level interaction pattern is as follows:

1. Club Governance defines capability structure.
2. Colleges onboard students and launch local chapters.
3. Students engage in structured certification pathways.
4. Companies access structured capability data.
5. Companies create internship/job requisitions.
6. Students apply through structured evaluation workflows.
7. Internship performance updates capability records.

8. Analytics engine updates readiness indicators.
9. Companies convert verified candidates to hires.

This forms a closed-loop ecosystem.

2.10.9 Control and Guardrail Architecture

The architecture enforces the following constraints:

1. Separation of Governance and Execution.

    2. Governance is centralized (Club Admin layer).

    3. Execution is distributed (Students and Colleges)

    4. Hiring authority is corporate-controlled but framework-aligned

    5. Skill Certification requires evidence validation

6. Mandatory approval workflows for:
    - Job requisitions
    - Internship requisitions
    - Certification upgrades
    - Corporate framework customizations
7. Evidence-backed certification logic.
8. Hierarchical role authority enforcement.
9. Audit trail logging for all evaluative decisions.
10. Version-controlled framework updates.

2.10.10 Scalability Model

The architecture supports:

- Multiple industries
- Multiple parent clubs per industry
- Multiple colleges per club
- Multiple companies per industry
- Thousands of students per institution

The system is horizontally scalable across:

- Users
- Institutions

- Corporate tenants
- Capability frameworks

2.10.11 Architectural Integrity Principle

The architectural design ensures:

- Governance remains centralized.
- Execution remains distributed.
- Evaluation remains evidence-driven.
- Hiring remains capability-mapped.
- No single actor can independently distort the system.

Whizard operates as a synchronized capability network rather than isolated applications. To summarize

Capabilities belong to Industry.
Competence belongs to Role.
Tasks prove Skills.
Evidence proves Tasks.
Supervisors certify Skills.
Capability Instance (All skills attached to it) completion define Role Readiness.
Role Competence defines Workforce Readiness.

This must now be consistently reflected across:

Certification logic

Internship logic

Employee promotion logic

AI learning path logic

3. Specific Requirements

3.1 Club Admin App

3.1.1 Introduction

The Club Admin App functions as the **Governance Layer** of the Whizard ecosystem.

It is responsible for:

- Industry and Club structure governance
- Workforce Readiness Capability Framework (WRCF) management
- Certification architecture definition
- Corporate onboarding
- Content provider onboarding
- Communication governance
- Corporate pricing and sponsorship configuration

No downstream application (Student, College, Company) can function without configuration performed within the Club Admin App.

3.1.2 Actors

The following actors operate within the Club Admin App:

1. System Admin
2. Club Admin
3. Club Corporate Manager
4. Club Content Manager
5. Club Communication Manager

Each role shall have strictly defined access privileges enforced by RBAC.

3.1.3 Functional Requirements

3.1.3.1 Parent Club Management

FR-CA-001
The system shall allow System Admin to create, edit, activate, or deactivate Parent Clubs.

FR-CA-002
Each Parent Club shall be mapped to an Industry domain.

FR-CA-003
Parent Club configuration shall include:

- Name
- Industry
- Description
- Governance settings
- Status (Active/Inactive)

FR-CA-004
Deactivation of a Parent Club shall not delete historical student or company data.

3.1.3.2 Club Management

FR-CA-005
The system shall allow Club Admin to create multiple Clubs under a Parent Club.

FR-CA-006
Each Club shall define:

- Certification levels (minimum 1, configurable maximum)
- Enrollment rules
- Assessment rules
- Completion criteria

FR-CA-007
The system shall enforce that a Club must belong to exactly one Parent Club.

FR-CA-008
Club Admin shall be able to define exploration rules (e.g., students may explore up to three clubs).

3.1.3.3 Workforce Readiness Capability Framework (WRCF) Management

FR-CA-009

Club Admin shall be able to define WRCF structure as per below.

FR-CA-NEW-001
Club Admin shall define Primary Work Objects (PWO) under Functional Groups.

FR-CA-NEW-002
Each PWO shall contain multiple Secondary Work Objects (SWO).

FR-CA-NEW-003
Capabilities shall be mapped to SWO.

FR-CA-NEW-004
Proficiency levels shall be configurable per Capability.

FR-CA-NEW-005
Each Skill shall contain:

Multiple Tasks

Associated Control Points

Evidence rules

FR-CA-010
Role Definitions shall assign multiple Capability Instances. A Capability Instance shall be defined as:
(SWO + Capability + Proficiency Level).

FR-CA-011
All framework modifications shall be version-controlled.

FR-CA-012
The system shall maintain a historical audit trail of framework changes.

FR-CA-013
Framework baseline definitions shall not be directly editable by Company-level users without version approval.

Sector, Industry, and Industry Reference Model Management

Req ID	Requirement	Priority	Acceptance criteria
FR-SEC-01	The platform shall allow Club Industry expert to create, update, and retire Sectors with descriptions and metadata.	Must	Sector CRUD available with audit trail; retired sectors cannot be selected for new industries.

FR-IND-01	The platform shall allow authorized Club Industry Expert to create and manage Industries under a Sector, including external taxonomy codes (e.g., NAICS/ISIC/GICS) as attributes.	Must	Industry can store multiple taxonomy mappings with source and effective date.
FR-IRM-01	The platform shall support Industry Reference Models (IRMs) as versioned templates for function groups, PWO, SWO, capabilities, Proficiencies, skills, task and evidences	Must	An IRM version can be created as draft, reviewed, and published as immutable; versions have status and changelog.
FR-IRM-02	The platform shall support creation of Company under Industry. Company admin than can create Departments and Roles under company. Roles shall be mapped to the Capability instances	Must	An IRM version can be created as draft, reviewed, and published as immutable; versions have status and changelog.
FR-IRM-03	The platform shall provide diff and impact analysis between IRM versions (added/changed/retired nodes and mappings).	Should	Diff report identifies affected roles, competencies, and mapped learning experiences.

Company Configuration via Inheritance

Req ID	Requirement	Priority	Acceptance criteria
FR-COMP-01	The platform shall support multi-tenant Companies, each linked to one Industry and one IRM version as baseline.	Must	Company creation requires selecting industry + IRM version; stored as references.
FR-COMP-02	The platform shall bootstrap a company's departments, and roles by instantiating the selected IRM version.	Must	After bootstrap, company has instance nodes with template references and correct relationships.
FR-COMP-03	Company administrators shall be able to override selected fields of inherited nodes (e.g., names, descriptions, owner metadata) without modifying the IRM.	Must	Override creates a tenant-level delta record; effective model reflects override; IRM remains unchanged.

FR-COMP-04	Company administrators shall be able to extend the model by adding new departments /roles / mapping of Capability instances that optionally reference existing templates or stand alone. role templates may be reused across industries.	Must	Extensions appear in the effective model and can participate in mappings and learning plans.
FR-COMP-05	The platform shall support company configuration export/import (JSON) for audit, migration, and backup.	Should	Export includes IRM version reference + tenant deltas; import restores equivalent effective configuration.

3.1.3.4 Certification Architecture Management

FR-CA-014
Club Admin shall define Certification Levels (e.g., Level 1 to Level 4).

FR-CA-015
Each Certification Level shall define:

- Required capabilities
- Required skill thresholds
- Required project completion
- Required assessment scores

FR-CA-016
The system shall enforce evidence-based certification advancement.

FR-CA-017
Certification logic shall automatically validate completion criteria before upgrade approval.

3.1.3.5 Corporate Onboarding & Management

FR-CA-018
Club Corporate Manager shall onboard Companies under a Parent Club.

FR-CA-019
Company onboarding shall capture:

- Company name
- Industry mapping

- Contact details
- Billing details
- Subscription package

FR-CA-020
The system shall allow configuration of corporate pricing plans.

FR-CA-021
The system shall track corporate sponsorships and annual subscriptions.

FR-CA-022
Deactivation of a company shall not delete historical internship or hiring records.

3.1.3.6 Content Provider Management

FR-CA-023
Club Content Manager shall onboard content service providers.

FR-CA-024
Service providers may be assigned:

- Courses
- Projects
- Training programs
- Events

FR-CA-025
The system shall allow assignment of content to specific certification levels.

FR-CA-026
Content modifications shall not invalidate historical student records.

3.1.3.7 Project & Event Governance

FR-CA-027
Club Admin shall create and manage industry-level projects.

FR-CA-028
Projects shall be mappable to specific capabilities and skills.

FR-CA-029
The system shall allow corporate-sponsored projects.

FR-CA-030
The system shall track project status (Draft, Active, Completed, Archived).

3.1.3.8 Communication Governance

FR-CA-031
Club Communication Manager shall moderate club-level social feeds.

FR-CA-032
The system shall allow role-scoped visibility of posts.

FR-CA-033
The system shall log moderation actions.

FR-CA-034
Communication privileges shall be club-bound and permission-driven.

3.1.3.9 Analytics & Reporting

FR-CA-035
The system shall provide dashboards for:

- Student enrollment per club
- Certification progression rates
- Corporate participation
- Internship conversion rates

FR-CA-036
Club Admin shall view industry-wide readiness trends.

3.1.3.10 Billing & Sponsorship Governance

FR-CA-037
The system shall allow creation of pricing tiers for corporate packages.

FR-CA-038
The system shall support invoicing and payment tracking.

FR-CA-039
Corporate advertisement campaigns shall be configurable with duration and scope.

3.1.3.11 Role-Based Access Control (RBAC)

FR-CA-040
The system shall enforce role-based access restrictions for all Club Admin users.

FR-CA-041
No Club-level user shall access data outside their Parent Club scope.

FR-CA-042
All administrative actions shall generate audit logs.

3.1.4 Data Integrity Requirements

1. All framework changes must be versioned.
2. Historical certification data must remain immutable.
3. Corporate sponsorship records must be auditable.
4. Deactivation shall not delete dependent records.
5. Student progress records must not be editable by Club Admin after certification approval.

3.1.5 Workflow Constraints

1. Certification upgrade requires:
    - Evidence validation
    - System verification of thresholds
    - Optional manual approval workflow
2. Company onboarding requires:
    - Subscription plan selection
    - Approval by Club Corporate Manager
3. Framework modification requires:
    - Draft state
    - Review state
    - Published state

3.1.6 Security Requirements (Club Admin App)

1. Multi-factor authentication for governance roles.
2. Role hierarchy enforcement.
3. Audit trail logging.
4. Data encryption in transit.
5. Tenant isolation enforcement.

3.1.7 Non-Functional Expectations (Club Admin App)

- High availability
- Framework update consistency
- Scalability across multiple industries
- Minimal latency for dashboard access
- Secure handling of corporate billing data

3.2 Student App

3.2.1 Introduction

The Student App functions as the **Execution and Signal Generation Layer** of the Whizard ecosystem.

It enables students to:

- Explore clubs
- Enroll in structured certification pathways
- Complete courses and projects
- Build capability evidence
- Participate in events
- Apply for internships and jobs
- Interact within Whizard Social
- Track readiness progression

All student actions must be capability-mapped and evidence-backed.

The Student App does not define standards — it operates within governance defined by the Club Admin App.

3.2.2 Actors

1. Student
2. Local Chapter Club Executive
3. Alumni

All permissions shall be governed via RBAC.

3.2.3 Functional Requirements

A. Student Onboarding & Profile Management

3.2.3.1 Student Registration

FR-SA-001
The system shall allow students to register using:

- Institutional email OR
- Verified onboarding by College Admin

FR-SA-002
The system shall capture:

- Personal details
- Academic year
- College
- Degree program
- Interests

FR-SA-003
The system shall map each student to exactly one College entity.

3.2.3.2 Student Profile

FR-SA-004
The system shall maintain a structured student profile including:

- Certifications
- Capability progression
- Projects completed
- Internship history
- Assessment scores
- Job applications

FR-SA-005
Students shall not be able to manually edit certified capability scores.

B. Club Exploration & Enrollment

3.2.3.3 Club Exploration Stage

FR-SA-006
Students shall be allowed to explore up to a configurable maximum number of clubs (default: 3).

FR-SA-007
Exploration access shall allow:

- Viewing club overview
- Viewing certification structure
- Viewing sample projects
- Viewing skill requirements

FR-SA-008
Exploration participation shall not count toward certification progression unless enrolled.

3.2.3.4 Primary Club Selection

FR-SA-009
Students shall be required to select one primary club for structured certification progression.

FR-SA-010
Once selected, switching primary clubs shall follow configurable rules.

FR-SA-011
The system shall enforce certification progression within one primary club at a time.

C. Certification Progression

3.2.3.5 Enrollment in Certification Level

FR-SA-012
Students shall enroll in Certification Level 1 upon joining primary club.

FR-SA-013
Enrollment in higher levels shall require completion validation of lower levels.

3.2.3.6 Certification Advancement

FR-SA-014
The system shall validate:

- Required capabilities
- Required skill thresholds
- Required projects
- Required assessments

before advancing a student to the next level.

FR-SA-015
Certification upgrades may require manual approval where configured.

FR-SA-016
Students shall receive notification upon certification advancement.

D. Courses & Learning Modules

3.2.3.7 Course Enrollment

FR-SA-017
Students shall enroll in courses mapped to their certification level.

FR-SA-018
The system shall track:

- Course progress
- Assessment scores
- Completion status

FR-SA-019
Course completion shall update mapped capability records.

E. Projects & Capability Evidence

3.2.3.8 Project Participation

FR-SA-020
Students shall enroll in projects aligned with certification requirements.

FR-SA-021
Projects shall be mapped to:

- Capabilities
- Skills
- Evidence types

FR-SA-022
Students shall submit deliverables via structured submission interface.

3.2.3.9 Project Evaluation

FR-SA-023
Project evaluations shall update:

- Skill scores
- Capability scores
- Evidence records

FR-SA-024
Students shall not modify evaluation results.

FR-SA-025
Project history shall remain permanently recorded.

F. Events & Contests

3.2.3.10 Event Participation

FR-SA-026
Students shall register for club events and contests.

FR-SA-027
Event participation may generate:

- Participation certificates
- Skill evidence
- Ranking metrics

G. Mentorship

3.2.3.11 Mentor Interaction

FR-SA-028
Students shall request mentorship sessions.

FR-SA-029
Mentorship sessions may be:

- Scheduled
- Logged
- Feedback-recorded

H. Internship & Job Applications

3.2.3.12 Internship Application

FR-SA-030
Students shall apply to internships for which they meet eligibility criteria.

FR-SA-031
Eligibility may depend on:

- Certification level
- Skill thresholds

- Capability requirements

FR-SA-032
The system shall track:

- Application status
- Shortlisting
- Interview stages
- Offer status

3.2.3.13 Internship Evaluation

FR-SA-NEW-001
Students shall be assigned Tasks by supervisor based on role wise Capability Instance.

FR-SA-NEW-002
Students shall upload Evidence per Task.

FR-SA-NEW-003
Evidence types may include:

Documents

Simulation logs

Field verification

Supervisor sign-off

Video/photo proof

FR-SA-NEW-004
Tasks shall be marked complete only after Evaluator approval.

FR-SA-NEW-005
Skill shall be marked achieved only after:

All tasks completed

All control points validated

Evidence approved

FR-SA-NEW-006
Capability Instance shall be marked achieved only after all Skills under it are certified.

FR-SA-NEW-007
Role Readiness shall be auto-computed when all assigned Capability Instances are certified.

FR-SA-033
Internship performance shall update:

- Capability evidence
- Skill progression

FR-SA-034
Internship records shall be visible to placement cell and authorized companies.

3.2.3.14 Job Application

FR-SA-035
Students shall apply for full-time roles.

FR-SA-036
Job eligibility shall be system-validated.

FR-SA-037
Offer acceptance shall be logged.

I. Job Readiness Index

3.2.3.15 Readiness Score

FR-SA-038
The system shall compute a Job Readiness Index (JRI) based on:

- Certification level
- Capability coverage
- Project depth
- Internship evaluations
- Assessment scores

FR-SA-039
Students shall view their JRI dashboard.

FR-SA-040
JRI calculation logic shall not be editable by students.

J. Whizard Social

3.2.4 Whizard Social Communities

3.2.4.1 Club-Based Communities

FR-SA-SOC-001
Students enrolled in a Club shall automatically join:

- **College-level club community**
- **Club community (same interest across India)**

FR-SA-SOC-002
Students shall view and interact with peers from other colleges enrolled in the same Club.

3.2.4.2 Community Content

Students may:

- **Share internship experiences**
- **Post project demos**
- **Share company interview insights**
- **Publish learning reflections**
- **Post college life experiences**
- **Share achievement milestones**

3.2.4.3 Peer-to-Peer Learning

FR-SA-SOC-003
Students shall:

- **Comment**
- **Like**
- **Collaborate**
- **Co-create project threads**
- **Form peer groups within club domain**

3.2.5 Official College Communication Platform

Whizard Social shall act as the official campus communication hub.

3.2.5.1 Broadcast Messaging

FR-SA-SOC-004
College Admin shall broadcast:

- **Placement updates**
- **Exam notifications**
- **Event announcements**
- **Internship drives**
- **Certification deadlines**

FR-SA-SOC-005
Students shall receive:

- **In-app notifications**
- **Push notifications**
- **Email (if enabled)**

3.2.5.2 Event Calendar

FR-SA-SOC-006
Students shall access a real-time gamified event calendar.

Calendar shall include:

- **Club events**
- **Trainings**
- **Internships**
- **Contests**
- **Placement drives**
- **Certification deadlines**

3.2.6 Engagement & Gamification Engine

This is critical because engagement drives the Employability Index indirectly.

3.2.6.1 Engagement Score

System shall compute Engagement Score based on:

- **Event participation**
- **Post interactions**
- **Peer collaboration**
- **Content contributions**
- **Challenge participation**

3.2.6.2 Social Contribution Scoring

Engagement shall contribute to Club Points under:

"Social Platform Activity"

Points shall follow weighted structure already defined in College App.

Participation → Completion → Excellence logic applies.

3.2.6.3 Challenges & Competitions

FR-SA-SOC-007
System shall host:

- **Weekly challenges**
- **National competitions**
- **Skill contests**
- **Hackathons**
- **Peer-voted demo showcases**

3.2.6.4 Showcase Profile

Students shall maintain a public profile including:

- **Club achievements**
- **Certifications**
- **Internship experiences**
- **Projects**
- **Contest wins**
- **Engagement highlights**

This becomes visible to:

- **Companies**
- **Mentors**
- **Peers**

3.2.7 Collaboration & Competition

3.2.7.1 Cross-College Collaboration

Students may:

- **Form project teams across colleges**
- **Collaborate in club challenges**
- **Co-submit innovation demos**

3.2.7.2 Peer Recognition

Students may:

- **Nominate peers for excellence**
- **Endorse project contributions**
- **Vote on best demo posts**

Excellence recognition contributes to "Excellence" component of Club Points.

3.2.8 Analytics & Feedback Engine

This is where Student App connects to College App intelligence.

3.2.8.1 Participation Tracking

System shall track:

- **Event attendance**
- **Certification attempts**
- **Task completion**
- **Internship engagement**
- **Social activity**

3.2.8.2 Skill Progress Dashboard

Students shall view:

- **Capability Instances completed**
- **Skills certified**
- **Tasks pending**
- **Role readiness %**

3.2.8.3 Engagement Heatmap

Students shall see:

- **Engagement vs peers**
- **Engagement percentile**

- **Growth recommendations**

3.2.8.4 AI-Powered Trending Topics

FR-SA-SOC-008
System shall use ML models to:

- **Surface trending discussion topics**
- **Highlight high-value internship threads**
- **Recommend relevant posts based on club interest**

3.2.9 Feedback & Placement Intelligence

3.2.9.1 Internship Experience Repository

Students shall:

- **Log internship experiences**
- **Upload insights**
- **Rate internship quality**
- **Share interview process details**

Collective intelligence grows over time.

3.2.9.2 Placement Preparation Signals

System shall:

- **Recommend trainings based on weak skill areas**
- **Alert student if Employability Index below branch threshold**
- **Suggest improvement path**

3.2.10 Community Moderation & Governance

3.2.10.1 Moderation Controls

- **ClubExec can moderate club-level posts.**
- **College Admin can moderate college-wide posts.**
- **Central Admin can moderate national threads.**

3.2.10.2 Anti-Spam & Integrity

- **Automated content filtering**

- **Abuse reporting**
- **Audit logs**

3.2.11 Employability Signal Amplification

Whizard Social amplifies:

- **Demonstrated excellence**
- **Skill mastery evidence**
- **Internship performance**
- **Peer recognition**

Student's public signal becomes:

Not just certification-based
But community-recognized competence-based.

3.2.12 Integration with Employability Index

Social & Engagement activity feeds:

- **Social Platform Activity score**
- **Engagement Score**
- **Excellence recognition score**
- **Contest performance metrics**

These update Club Points → Employability Index → College Ranking.

Strategic Upgrade Summary

Student App now operates as:

- **Skill Execution Layer**
- **Social Capital Builder**
- **Official Campus Communication Channel**
- **Engagement Scoring Engine**
- **Internship Knowledge Repository**
- **Employability Signal Amplifier**
- **National Peer Network**

K. Local Chapter Club Executive Privileges

3.2.14 Chapter Management

FR-SA-044
Local Chapter Club Executives may:

- Coordinate events
- Approve local participation
- Communicate with club members

FR-SA-045
Chapter Executives shall not modify certification logic.

L. Alumni Access

3.2.15 Alumni Permissions

FR-SA-046
Alumni shall:

- Access historical records
- Participate in mentorship
- Engage in Whizard Social

FR-SA-047
Alumni shall not enroll in new certification pathways.

3.2.4 Data Integrity Requirements

1. Certification records must be immutable once awarded.
2. Project evaluation records must be auditable.
3. Internship ratings must be preserved permanently.
4. Students cannot edit skill scores.
5. Historical club participation must be retained.

3.2.5 Workflow Constraints

1. Certification advancement must be rule-validated.
2. Internship eligibility must be system-verified.
3. Job eligibility must be threshold-based.

4. Evidence must exist before capability score updates.
5. Switching primary club must follow governance rules.

3.2.6 Security Requirements (Student App)

1. Authentication required for all actions.
2. Students may access only their own records.
3. Role escalation prohibited.
4. All submissions must be timestamped.
5. Application data must be encrypted in transit.

3.2.7 Non-Functional Requirements (Student App)

- High concurrency support
- Real-time progress updates
- Low-latency dashboard loading
- Secure document upload handling
- Scalable storage for project submissions

3.3 College App

3.3.1 Introduction

The College App functions as the **Institutional Integration and Monitoring Layer** of the Whizard ecosystem.

It enables colleges to:

- Onboard students
- Launch and manage local club chapters
- Upload academic transcripts
- Monitor student progression
- View job readiness analytics
- Coordinate internships with companies
- Track placement pipelines

The College App does not define capability standards.
It operates within the governance defined by the Club Admin App and leverages student data generated through the Student App.

Actors

1. College Admin
2. Placement Cell
3. Club Coordinator

Access shall be governed through RBAC.

3.3.2 Functional Requirements

A. College Onboarding & Profile Management

3.3.2.1 College Registration

FR-CO-001
The system shall allow onboarding of a College by Club Admin

FR-CO-002
College profile shall include:

- Institution name

- Accreditation details
- Contact information
- Authorized administrators
- Program offerings

FR-CO-003
A College shall operate as a distinct tenant within the platform.

3.3.3 Student Onboarding & Management

3.3.3.1 Student Registration

FR-CO-001
College Admin shall onboard students individually or via bulk upload.

The system shall validate uploaded data for:

- Duplicate entries
- Invalid identifiers
- Missing mandatory fields

FR-CO-002
Student attributes shall include:

- Stream
- Branch
- Academic Year
- Contact details
- Enrollment status
- Login status

FR-CO-003
System shall track onboarding status:

- Not Joined
- Joined
- Logged In
- Onboarded

3.3.3.2 Student Status Tracking

FR-CO-004
College Admin shall filter students by:

- Stream
- Branch
- Year
- Status

FR-CO-005
System shall maintain full onboarding audit logs.

Transcript Upload

College Admin shall upload semester-wise academic transcripts.

Transcript data may include:

- GPA/CGPA
- Subject-level scores
- Backlog indicators

Transcript data shall be stored as structured academic records. Transcript uploads shall not override student capability scores.

3.3.4 Club Onboarding & Governance

3.3.4.1 Club Chapter Creation

FR-CO-006
College Admin shall launch local chapters mapped to Parent Clubs.

FR-CO-007
Each chapter shall assign:

- President
- Vice Presidents
- Secretary
- Treasurer

3.3.4.2 Representative Governance

FR-CO-008
System shall maintain club office bearer records per academic cycle.

FR-CO-009
Performance of club leadership shall reflect in club-level performance analytics.

3.3.5 Employability Index & Club Points Engine

This is the core upgrade.

3.3.5.1 Employability Index Definition

Student Employability Index shall be computed based on:

Weighted Club Points earned across activities.

Weightage Order (High → Low):

1. Job Trainings
2. Internships
3. Events
4. Certification Levels
    - Assessments
    - Projects
    - Courses
5. Club Exploration & Joining
6. Social Platform Activity

3.3.5.2 Club Points Composition

For each activity, points shall be computed using:

Where:

- Participation (Initiation) = 10%
- Completion (Effort) = 40%
- Excellence (Demonstrated & Recognized Skill) = 50%

3.3.5.3 Activity Weight Governance

FR-CO-010
Activity weights shall be configurable at system level.

FR-CO-011
Higher weighted activities (e.g., Job Training, Internship) shall contribute disproportionately to the Employability Index.

3.3.5.4 Student Employability Index Calculation

FR-CO-012
System shall compute:

- Branch-level average Employability Index
- Year-level distribution
- College-level index score

3.3.6 Student Progress Tracking

3.3.6.1 Individual Student Dashboard

System shall display:

- Club Score
- Certification Status
- Training Participation
- Internship Status
- Final Selection Status
- Employability Tier:
    - Achiever
    - On Track
    - Growth Needed

3.3.6.2 Branch-wise Performance

FR-CO-013
College shall view branch-wise:

- Training selection count
- Placement offers
- Certification levels
- Average branch score

3.3.7 Placement Dashboard & Funnel Analytics

3.3.7.1 Placement Metrics

System shall track:

- Total Students
- Students Placed
- Students Pending
- Placement Rate
- Internship Placement Rate
- Paid Internship %

- Average Salary
- Highest Salary

3.3.7.2 Placement Funnel

FR-CO-014
System shall display Journey to Placement Funnel:

Stages:

1. Profile Completed
2. Internship Done
3. Training Attended
4. Drive Applied
5. Placement

Drop-off percentage shall be calculated at each stage.

3.3.7.3 Placement by Salary Range

System shall categorize placements by:

- Salary slabs
- Industry type
- Internship vs Job

Internship Mid-term feedback Coordination

Functional Managers from companies shall submit internship feedback.
Placement Cell shall access mid-term and final evaluation reports.
Feedback shall update student capability evidence automatically.

3.3.8 Club Performance Dashboard

3.3.8.1 Club Enrollment Metrics

System shall display:

- Total Clubs
- Enrollment Rate
- Engagement Score
- Achievers %
- On Track %
- Room for Growth %

3.3.8.2 Club Points Screen

For each club:

System shall show:

- Avg Engagement Score
- Avg Club Points
- Avg Certification Score
- Event Scores
- Social Engagement
- Exploration Score

3.3.8.3 Individual Club Student Listing

System shall show:

- Login Status
- Engagement Score
- Club Points
- Social Activity
- Event Participation
- Certification Status

3.3.9 Student Progress Categorization

System shall classify students:

- Achiever (High Employability Index)
- On Track
- Growth Needed

Thresholds configurable.

3.3.10 College Ranking Engine

3.3.10.1 Ranking Metrics

College Ranking shall consider:

- Placement Rate
- Internship Completion
- Employability Index Average

- Certification Levels Achieved
- Club Engagement Rate
- High Achiever %

3.3.10.2 Multi-Level Ranking

System shall support:

- District Ranking
- State Ranking
- Zone Ranking
- National Ranking

3.3.10.3 Performance Breakdown

College performance areas:

- Club Enrollment
- High Achievers
- On Track %
- Growth Needed %
- Placement Rate
- Internship Completion %

3.3.11 Placement Logbook Access

FR-CO-015
Placement Cell shall access student internship logbooks.

FR-CO-016
Placement Cell shall view company-wise internship outcomes.

3.3.12 Predictive Placement Analytics

FR-CO-017
System shall predict:

- Expected placement rate
- Branch risk indicators
- Students at placement risk
- Certification gap impact on placement probability

3.3.13. Communication & Notifications

3.3.13.1 College Communication

FR-CO-018
College Admin shall broadcast announcements to:

- All students
- Specific club members
- Specific academic year groups

FR-CO-019
College-level posts shall be visible only within the institution.

3.3.14 Corporate Interaction Visibility

3.3.14.1 Company Engagement Overview

FR-CO-020
Placement Cell shall view:

- Companies tied to the College
- Active internships
- Active job requisitions
- Historical hiring trends

3.3.14.2 Student Application Visibility

FR-CO-021
Placement Cell shall view application statuses of students.

FR-CO-022
College users shall not interfere with company hiring decisions.

3.3.12 Data Integrity & Governance

1. Club Points once awarded cannot be modified without audit trail.
2. Excellence score must be tied to certified achievement.
3. Placement data must be verified via company confirmation.
4. Ranking computations must be version-controlled.
5. Transcript uploads must preserve version history.
6. Internship feedback must remain immutable once submitted.
7. College-level analytics must be read-only derived data.
8. Student certification records must not be editable by college users.

9. Student academic records must be auditable.

3.3.4 Structural Summary

College App operates as:

- Student onboarding system
- Club governance platform
- Employability computation engine
- Placement intelligence dashboard
- Ranking & benchmarking system
- Predictive institutional analytics layer

3.3.5 Workflow Constraints

1. Student onboarding requires data validation.
2. Local chapter creation requires Club-level approval.
3. Internship feedback must follow company submission.
4. Job readiness index is system-calculated and read-only.
5. Placement Cell visibility is role-restricted.

3.3.6 Security Requirements (College App)

1. Institutional data isolation.
2. Role-based access within college.
3. Secure bulk upload validation.
4. Audit trail logging for:
    - Student onboarding
    - Transcript uploads
    - Chapter creation
5. No cross-college data visibility.

3.3.7 Non-Functional Requirements (College App)

- Scalable student data handling
- Real-time analytics dashboard refresh
- Secure document storage
- High availability during placement season
- Performance optimized for bulk data upload

3.4 Company App

3.4.1 Introduction

The Company App functions as the **Industry Integration and Talent Absorption Layer** of the Whizard ecosystem.

It enables companies to:

- Onboard and manage organizational structure
- Customize capability requirements aligned to industry baseline
- Post internships and job requisitions
- Evaluate interns and candidates
- Sponsor projects and events
- Run advertisement campaigns
- Provide conditional offer training
- Track hiring analytics
- Manage billing and subscriptions

It is designed to:

- Replace or significantly augment traditional L&D systems
- Reduce hiring risk through capability-verifiable evaluation
- Convert learning spend into measurable workforce readiness
- Provide AI-powered skill intelligence
- Build a continuous college-to-enterprise talent pipeline
- Serve as an immediate go-to resource for company-specific problem solving

Traditional systems (LMS, generic courses, interview-based hiring) produce low-confidence outcomes. Whizard instead operates on:

- Workforce Readiness Capability Framework (WRCF)
- AI-driven skill intelligence
- Multi-agent simulations
- Work-object mastery validation
- Evidence-based capability scoring

The Company App operates within capability frameworks governed by the Club Admin App and interacts with student-generated performance data.

3.4.2 Enterprise Capability Architecture

The Company App extends the WRCF into enterprise execution.

It allows companies to:

- Map internal roles to industry roles
- Define role competence thresholds
- Track workforce capability coverage
- Identify skill gaps at organization level
- Drive AI-personalized upskilling
- Validate learning via work-object simulation

3.4.2.1 Actors

1. Company Admin
2. HR Head
3. Recruitment Manager
4. Functional Manager
5. Training Manager
6. Department Head
7. Department Manager

All permissions shall be governed via RBAC.

3.4.3 Functional Requirements

A. Company Onboarding & Organizational Setup

3.4.3.1 Company Registration

FR-CM-001
The system shall allow onboarding of a Company by club Admin under the respective Industry.

FR-CM-002
Company profile shall include:

- Company name
- Industry
- Registered address
- Billing details

- Subscription package
- Authorized user (company Admin details)

FR-CM-003
Each company shall operate as an isolated tenant within the platform.

3.4.3.2 Organizational Structure Setup

FR-CM-004
Company Admin shall map Departments and Role Positions to WRCF baseline Role Definitions.

FR-CM-005
Departments shall be mapped to Functional Groups defined in the Workforce Readiness Capability Framework (WRCF).

FR-CM-006
Company Admin shall create internal users and assign role-based permissions.

B. Industry Reference Models and Company Inheritance

Industry Reference Models and Company Inheritance

To avoid configuring each company separately, the platform must support Industry Reference Models (IRMs): versioned templates that define a generic WRCF for an industry. Companies instantiate an IRM version to produce a tenant-specific configuration, then optionally override or extend elements while retaining lineage.

Company Instantiation and Override Model

- Instantiation creates Company-specific instances that inherit from template nodes.
- Three change types are supported: Adopt (no change), Override (modify fields), Extend (add new child nodes).
- Overrides do not mutate the IRM; they apply at tenant scope only and maintain a reference to the template source.
- Effective configuration is computed by merging IRM + tenant overrides + extensions with deterministic precedence (tenant > IRM).

Inheritance and Override Rules

1. Template nodes are read-only for tenant users; only curators can modify IRM drafts and publish versions.
2. Tenant overrides must retain template references and cannot violate core constraints (e.g., cannot change entity type).

3. Deletion of inherited nodes is expressed as a tenant-level 'retire' flag with reason and audit trail (soft delete), not physical removal.
4. Upgrading IRM versions must never auto-delete tenant-created extensions; conflicts are resolved through explicit mapping or duplication with lineage.

3.4.3.3 Role Mapping

FR-CM-007
Company Admin shall define company-specific Role Positions.

FR-CM-008
Each Role Position shall map to baseline Industry Roles from WRCF.

FR-CM-009
Company Admin may add additional skill requirements without altering industry baseline definitions.

Role Competence Engine

FR-CM-NEW-008
Role shall define required Capability Instances and may need to customize the industry baseline by adding / deleting certain Capability Instances.

FR-CM-NEW-009
System shall track per employee:

- Skills completed (as per the Capability instances assigned for the role)
- Pending tasks
- Evidence status
- Role readiness percentage

FR-CM-NEW-010
Employee becomes “Role Competent” only when all capability instances are certified.

3.4.3.4 Capability Customization

FR-CM-010
Company may:

- Adjust required skill thresholds
- Add company-specific competencies
- Define minimum certification level eligibility

FR-CM-011
Customizations shall be version-controlled and auditable. Baseline Industry definitions shall not be editable by Company users.

3.4.3.5 Enterprise Skill Gap Identification

FR-CM-012
The system shall analyze employee capability data to identify:

- Skill gaps
- Role readiness gaps
- Promotion readiness indicators
- Organizational risk areas

The system shall provide department-level skill gap dashboards

C. Internship Management

3.4.3.5 Internship Requisition Creation

FR-CM-013
Department Manager shall create Internship Requisitions.

FR-CM-014
Internship requisition shall define:

- Role position
- Required certification level
- Required capability thresholds
- Internship duration
- Location (On-site/Hybrid/Remote)
- Stipend details
- Week wise Skills and Task assignment as per the capability instances attached to the Role.

3.4.3.6 Internship Approval Workflow

FR-CM-015
Internship requisition shall require approval by:

- Department Head
- HR Head (if configured)

FR-CM-016
Only approved requisitions shall be visible to eligible students.

3.4.3.7 Application Management

FR-CM-017
Recruitment Manager shall:

- View eligible applicants
- Shortlist candidates
- Schedule interviews
- Record interview outcomes

FR-CM-018
Application statuses shall include:

- Applied
- Shortlisted
- Interview Scheduled
- Selected
- Rejected
- Offer Extended
- Offer Accepted

3.4.3.8 Internship Evaluation

FR-CM-019
Functional Manager shall submit:

- Mid-term feedback
- Final evaluation
- Capability ratings
- Internship performance score

FR-CM-020
Evaluators shall review submitted evidence against Task / Control Points.

FR-CM-NEW-005
Evaluator shall:

Approve

Reject

Request revision

FR-CM-NEW-006
Approval shall update Skill certification status.

FR-CM-NEW-007
The system shall maintain evaluator identity and timestamp.

FR-CM-021
Internship evaluation once submitted shall be immutable.

D. Job Management

3.4.3.9 Job Requisition Creation

FR-CM-022
Department Manager shall create Job Requisitions.

FR-CM-023
Job requisitions shall define:

- Role position
- Required certification level
- Required capability coverage
- Compensation range
- Location

3.4.3.10 Job Approval Workflow

FR-CM-024
Job requisitions shall require:

- Department Head approval
- HR Head approval

3.4.3.11 Candidate Selection

FR-CM-025
Recruitment Manager shall manage:

- Shortlisting

- Interview scheduling
- Interview feedback logging
- Offer extension

FR-CM-026
Offer acceptance shall be recorded in the system.

E. Training & Conditional Offers

3.4.3.12 Conditional Offer Training

FR-CM-027
Training Manager shall create:

- Training programs
- Assignments
- Assessments

FR-CM-028
Training may be assigned to:

- Interns
- Conditional offer candidates
- Existing employees

3.4.3.13 Training Evaluation

FR-CM-029
Training completion shall generate:

- Assessment score
- Competency rating
- Certification (if applicable)

F. AI-Powered Personalized Learning & Reskilling

3.4.3.14 Personalized Learning Path Generation

FR-CM-030
AI engine shall generate personalized learning paths based on:

- Current capability score
- Target role requirements

- Career trajectory
- Performance data

FR-CM-031
Learning paths shall dynamically adjust based on assessment outcomes.

3.4.3.15 Dynamic Content Creation

FR-CM-032
AI shall assist in generating:

- Refresher training modules
- Scenario-based simulations
- Role-specific micro-learning units
- Assessment questions
- Case studies

FR-CM-033
Generated content shall be tagged to capabilities and skills.

3.4.3.16 Work-Object Simulation

FR-CM-034
The system shall simulate real company work objects (e.g., troubleshooting scenarios, operational cases).

FR-CM-035
Simulation outcomes shall generate verifiable evidence of mastery.

This replaces passive LMS consumption with active capability validation.

G. Refresher & On-Demand Knowledge System

3.4.3.17 Enterprise Problem-Solving AI Assistant

FR-CM-036
Employees shall query company-specific knowledge through AI-assistant.

FR-CM-037
The assistant shall provide:

- SOP references
- Troubleshooting flows
- Best practices

- Capability-linked guidance

3.4.3.7 Continuous Refresher Training

FR-CM-038
Training Manager shall assign refresher programs mapped to:

- Capability decay risk
- Regulatory updates
- Process changes

FR-CM-039
Completion shall update capability freshness indicators.

H. Hiring Risk Reduction

3.4.3.19 Capability-Based Candidate Evaluation

FR-CM-040
Hiring decisions shall be based on:

- Verified certification levels
- Capability coverage
- Internship simulation results
- Work-object evidence

FR-CM-041
AI shall generate capability gap reports for shortlisted candidates.

3.4.3.20 Internship as Predictive Hiring Model

FR-CM-042
Internship evaluations shall map directly to the role readiness index.

FR-CM-043
System shall calculate Internship-to-Role Predictive Score.

I. Continuous Talent Pipeline

3.4.3.21 College-to-Enterprise Pipeline Tracking

FR-CM-044
Companies shall track:

- Talent pool pipeline by certification level
- College-wise readiness distribution
- Future hiring pipeline visibility

3.4.3.22 Early Capability Engagement

FR-CM-045
Companies may sponsor:

- Capability-aligned challenges
- Simulation competitions
- Problem-based events

This ensures early alignment of student capability to company need.

J. Project Sponsorship

3.4.3.23 Sponsored Projects

FR-CM-046
Recruitment Manager shall sponsor projects under Clubs.

FR-CM-047
Sponsored projects shall be mapped to defined capabilities.

FR-CM-048
The company shall receive performance analytics of participating students.

K. Advertisement & Branding

3.4.3.24 Ad Campaign Management

FR-CM-049
Company Admin shall create advertisement campaigns & sponsorships.

Create / Manage Sponsorship: Options to sponsor a skill club event, place an ad on the platform, or highlight a brand message.

Create New Campaign leads to a form for specifying sponsorship type, event details (if sponsoring a real-world event), ad creative uploads, and budget details. Campaign List shows each sponsorship or ad item, with performance metrics (impressions, clicks, or conversions).

Ad Campaign Dashboard: Track impressions, clicks, and overall ROI for ads placed on the platform.

FR-CM-050
Ads may target:

- Specific Clubs
- Specific Certification Levels
- Specific Colleges

FR-CM-051
Campaign duration and billing shall be configurable.

L. Analytics & Talent Insights

3.4.3.25 Talent Pool Access

FR-CM-052
Companies shall access filtered talent pools based on:

- Certification level
- Capability coverage
- College
- Internship ratings

3.4.3.26 Hiring Analytics Dashboard

FR-CM-053
System shall provide:

- Application volume trends
- Offer conversion rates
- Internship-to-full-time conversion rate
- Capability gap analysis

3.4.3.27 Workforce Intelligence Dashboard

System shall provide:

- Department capability heatmaps
- Role readiness index
- Promotion readiness signals
- Skill decay alerts
- Training ROI metrics
- Hiring risk score

M. Billing & Subscription Management

3.4.3.28 Subscription Management

FR-CM-054

Company Admin shall manage subscription plans.

Package Selection / Purchase: Allows the recruitment manager to purchase or upgrade packages for posting credits, priority placement, or sponsorships.

Payment History: Overview of all payments, invoices, usage of purchased credits.

Current Plan and usage (e.g., 5 postings used out of 10 available).

Available Packages

- ☐ **RPO** (Recruitment process Outsourcing) Plan packages with tiered pricing
- ☐ **Training Plan** (Inclusive of RPO)
- ☐ **Add-on Plans**: Enhanced Brand Promotion (while will provide additional opportunity to campaign over Social and/or Project / Events / Club sponsorship.

FR-CM-055

System shall generate invoices for:

- Annual subscription
- Sponsored projects
- Ad campaigns

FR-CM-056

Payment status shall be tracked.

N. Company Page Management

Individuals Profile Settings: Recruitment manager Uploads his profile pic, Bio, detailed profile and role at the current company,

Company Profile Settings: Upload logo, brand assets, company summary, mission/vision.

Content Management: Post articles, short updates, success stories, or upcoming events.

Engagement Metrics: Track visits, follower count, candidate engagement.

Edit Company Profile: upload logo/banner, set tagline, add a description or mission statement.

Post Article: share news, events, success stories; can appear on the company page or across the platform.

Manage Gallery: control images or media that appear to showcase your brand.

O. Data Governance & Restrictions

3.4.3.29 Access Control

FR-CM-057
Company users shall access only their organization's data.

FR-CM-058
Company users shall not modify:

- Industry baseline framework
- Student evaluation records outside their internships

3.4.4 Data Integrity Requirements

1. Internship evaluations must be immutable.
2. Hiring records must be auditable.
3. Framework customizations must be versioned.
4. Offer records must be permanently stored.
5. Billing records must comply with financial audit standards.

3.4.5 Workflow Constraints

1. Internship and job requisitions must follow approval workflow.
2. Eligibility validation must be automated.
3. Capability updates must be evidence-driven.
4. Training evaluation must be score-based.
5. No hiring status may bypass defined states.

3.4.6 Security Requirements (Company App)

1. Role-based permission hierarchy.
2. Secure candidate data handling.
3. Multi-factor authentication for HR Head.
4. Audit logs for hiring decisions.
5. Encrypted billing data.

3.4.7 Non-Functional Requirements (Company App)

- High availability during campus hiring season.
- Secure document upload for resumes and evaluations.
- Fast candidate filtering performance.
- Scalable analytics engine.
- Secure financial transaction handling.

4. External Interface Requirements

4.1 Introduction

This section defines all external interface requirements for the Whizard platform.

Whizard shall support integration with third-party systems to enable:

- Communication
- Video collaboration
- Artificial intelligence services
- Enterprise authentication
- Professional profile verification
- Course content delivery

All integrations shall be managed through the Whizard API Gateway and shall comply with security, privacy, and data isolation requirements.

4.2 System Interface Requirements

Whizard shall expose and consume RESTful APIs via a centralized API Gateway.

4.2.1 API Gateway Requirements

ER-SI-001
The system shall expose secure API endpoints for third-party integrations.

ER-SI-002
All external integrations shall route through the API Gateway.

ER-SI-003
The API Gateway shall enforce:

- Authentication
- Authorization
- Rate limiting
- Request validation
- Logging

ER-SI-004
All API communication shall use HTTPS with TLS encryption.

4.3 Communication Interfaces

4.3.1 Email Integration

ER-COM-001
The system shall integrate with third-party email service providers (e.g., SendGrid, SES, etc.).

ER-COM-002
Email notifications shall support:

- Account activation
- Certification updates
- Internship application updates
- Job offer notifications
- Billing notifications

ER-COM-003
Email templates shall be configurable.

4.3.2 SMS Integration

ER-COM-004
The system shall integrate with SMS gateway providers.

ER-COM-005
SMS notifications shall support:

- OTP verification
- Interview reminders
- Offer alerts

ER-COM-006
SMS delivery logs shall be stored for audit purposes.

4.3.3 In-App Notifications

ER-COM-007
The system shall provide real-time in-app notifications.

ER-COM-008
Notification preferences shall be configurable per user.

4.4 Video Conferencing Integration

Whizard shall integrate with external video conferencing platforms.

Supported platforms may include:

- Zoom
- Google Meet
- Microsoft Teams

4.4.1 Meeting Creation

ER-VC-001
Authorized users shall create meeting links directly from Whizard.

ER-VC-002
The system shall generate meeting invitations with:

- Date
- Time
- Participants
- Role context (Interview / Mentorship / Training)

4.4.2 Authentication & Token Handling

ER-VC-003
Integration shall use OAuth 2.0 authentication.

ER-VC-004
Meeting tokens shall not be stored in plaintext.

4.4.3 Meeting Metadata

ER-VC-005
Meeting metadata (link, time, participants) shall be stored within Whizard.

ER-VC-006
The system shall not store video recordings unless explicitly configured.

4.5 AI API Integrations

Whizard shall integrate with AI service providers including:

- OpenAI (ChatGPT APIs)
- Anthropic (Claude / Opus APIs)
- Future AI provider APIs

4.5.1 AI Use Cases

AI integrations may support:

- Resume analysis
- Capability gap analysis
- Interview question generation
- Skill assessment generation
- Certification pathway recommendations
- Content summarization
- Automated evaluation assistance

4.5.2 API Usage Requirements

ER-AI-001
All AI API calls shall pass through backend servers (no direct client calls).

ER-AI-002
Sensitive student data shall be masked or minimized before transmission.

ER-AI-003
AI outputs shall not automatically override human evaluation decisions.

ER-AI-004
AI recommendations shall be marked as advisory.

4.5.3 Data Privacy Guardrails

ER-AI-005
Personally identifiable information (PII) shall not be sent unless necessary.

ER-AI-006
AI response logs shall be stored securely for audit where applicable.

4.6 Single Sign-On (SSO) Integration

Whizard shall support enterprise authentication integrations.

4.6.1 Corporate HRMS Integration

ER-SSO-001
The system shall support SSO using:

- SAML 2.0
- OAuth 2.0
- OpenID Connect

ER-SSO-002
Corporate users shall authenticate using their HRMS credentials.

ER-SSO-003
Role mapping shall be configurable during SSO onboarding.

4.6.2 College ERP Integration

ER-SSO-004
Colleges shall integrate via SSO with ERP systems.

ER-SSO-005
Student identity verification shall support:

- ERP ID mapping
- Batch synchronization

4.6.3 Provisioning & De-Provisioning

ER-SSO-006
The system shall support automated user provisioning.

ER-SSO-007
Deactivated ERP/HRMS users shall lose Whizard access automatically.

4.7 LinkedIn Integration

Whizard shall integrate with LinkedIn APIs (subject to LinkedIn API policy compliance).

4.7.1 Mentor Profile Import

ER-LI-001
Mentors may connect LinkedIn accounts to import:

- Name
- Designation
- Company
- Experience summary
- Profile photo

4.7.2 Verification

ER-LI-002
LinkedIn data shall be used for informational purposes only.

ER-LI-003
Profile updates shall require user consent.

4.7.3 Data Storage

ER-LI-004
Imported LinkedIn data shall not be redistributed outside the platform.

4.8 Course Content Integration

Whizard shall integrate with external learning platforms.

Examples:

- LMS platforms
- MOOC providers
- SCORM-compliant systems
- Proprietary content providers

4.8.1 Content Access

ER-CC-001
Students may access external course content via secure redirection or embedded modules.

ER-CC-002
The system shall track:

- Course start
- Progress
- Completion

4.8.2 Content Standards

ER-CC-003
Integration shall support:

- SCORM
- xAPI

- LTI (Learning Tools Interoperability)

4.8.3 Completion Sync

ER-CC-004
External completion data shall update capability records automatically.

ER-CC-005
Course completion data must be verifiable and auditable.

4.9 Security & Compliance Requirements

All external integrations shall comply with:

1. TLS encryption
2. OAuth 2.0 / SAML authentication
3. API rate limiting
4. Data masking where required
5. Audit logging
6. Tenant data isolation

4.10 Failure Handling & Resilience

ER-FH-001
External service failure shall not cause system-wide outage.

ER-FH-002
Retry mechanisms shall be implemented for transient failures.

ER-FH-003
Users shall receive graceful error notifications.

4.11 Performance Requirements

1. External API response timeout thresholds must be defined.
2. AI API latency must not block critical workflows.
3. Notification dispatch must occur within defined SLA.
4. SSO login latency must be within acceptable enterprise standards.

4.12 Integration Governance Principles

1. All integrations must pass through the API Gateway.

2. No client-side direct third-party API calls.
3. No uncontrolled PII sharing.
4. AI outputs are advisory, not authoritative.
5. Corporate SSO must respect tenant isolation.

5. Non-Functional Requirements

5.1 Introduction

This section defines the non-functional requirements (NFRs) for the Whizard platform.

These requirements govern:

- Performance
- Scalability
- Security
- Availability
- Reliability
- Compliance
- Maintainability
- Auditability

These requirements apply across all four applications:

- Club Admin App
- Student App
- College App
- Company App

5.2 Performance Requirements

5.2.1 Response Time

NFR-PERF-001
The system shall respond to standard user interactions within ≤ 2 seconds under normal load.

NFR-PERF-002
Dashboard loading time shall not exceed 3 seconds for standard datasets.

NFR-PERF-003
Search and filtering operations shall return results within ≤ 2 seconds for datasets up to defined thresholds.

5.2.2 Concurrent Users

NFR-PERF-004
The system shall support concurrent usage by:

- ≥ 10,000 active students per tenant
- ≥ 1,000 concurrent internship/job applications during peak placement cycles

NFR-PERF-005
The system shall handle peak loads during:

- Certification result announcements
- Campus hiring seasons
- Internship application deadlines

5.2.3 AI & External API Latency

NFR-PERF-006
AI-assisted workflows shall not block critical system operations.

NFR-PERF-007
External API timeouts shall be configurable.

5.3 Scalability Requirements

5.3.1 Horizontal Scalability

NFR-SCAL-001
The system architecture shall support horizontal scaling of:

- Application servers
- API gateways
- Database clusters

5.3.2 Multi-Tenant Scalability

NFR-SCAL-002
The platform shall support onboarding of:

- Multiple industries
- Multiple Parent Clubs per industry
- Multiple colleges per club
- Multiple companies per industry

NFR-SCAL-003
Tenant data isolation shall not degrade performance across tenants.

5.3.3 Storage Scalability

NFR-SCAL-004
The system shall support scalable storage for:

- Project submissions
- Internship reports
- Course materials
- Transcript uploads

5.4 Availability Requirements

5.4.1 Uptime

NFR-AV-001
The system shall maintain ≥ 99.5% uptime annually.

NFR-AV-002
Planned maintenance windows shall be communicated in advance.

5.4.2 Fault Tolerance

NFR-AV-003
The system shall implement redundancy for:

- Application servers
- Database clusters
- Load balancers

NFR-AV-004
Failure of external integrations shall not cause system-wide outage.

5.4.3 Backup & Disaster Recovery

NFR-AV-005
Automated daily backups shall be maintained.

NFR-AV-006
Recovery Point Objective (RPO) shall not exceed 24 hours.

NFR-AV-007
Recovery Time Objective (RTO) shall be defined and documented.

5.5 Security Requirements

Security is a critical requirement due to:

- Student personal data
- Corporate hiring data
- Academic transcripts
- Financial billing records

5.5.1 Authentication & Authorization

NFR-SEC-001
All users shall authenticate using secure login mechanisms.

NFR-SEC-002
Multi-Factor Authentication (MFA) shall be mandatory for:

- System Admin
- Club Admin
- HR Head
- Company Admin
- College Admin

NFR-SEC-003
Role-Based Access Control (RBAC) shall be enforced across all modules.

5.5.2 Data Protection

NFR-SEC-004
All data in transit shall use TLS encryption.

NFR-SEC-005
Sensitive data at rest shall be encrypted.

NFR-SEC-006
Personally Identifiable Information (PII) shall be access-controlled.

5.5.3 Audit & Logging

NFR-SEC-007
The system shall log:

- Certification upgrades

- Internship evaluations
- Hiring decisions
- Framework modifications
- Billing transactions

NFR-SEC-008
Audit logs shall be tamper-resistant.

5.5.4 Session Management

NFR-SEC-009
User sessions shall expire after configurable inactivity periods.

NFR-SEC-010
Concurrent session control shall be configurable.

5.6 Data Privacy & Compliance

5.6.1 Regulatory Compliance

NFR-COMP-001
The system shall comply with applicable data protection regulations (e.g., GDPR-equivalent frameworks where applicable).

NFR-COMP-002
User consent shall be obtained before:

- Sharing student data with companies
- Importing LinkedIn data
- Sending promotional communications

5.6.2 Data Retention

NFR-COMP-003
Data retention policies shall be configurable.

NFR-COMP-004
Historical certification and internship records shall not be deleted.

5.6.3 Data Minimization

NFR-COMP-005
Only required data fields shall be transmitted to external APIs.

5.7 Reliability Requirements

NFR-REL-001
Critical workflows (internship application, certification advancement) shall be transactionally safe.

NFR-REL-002
System shall prevent partial state updates during failure scenarios.

NFR-REL-003
Retry mechanisms shall exist for transient external failures.

5.8 Usability Requirements

NFR-USAB-001
The system shall provide intuitive dashboards tailored to user roles.

NFR-USAB-002
Navigation shall not exceed three clicks for critical workflows.

NFR-USAB-003
Error messages shall be descriptive and actionable.

5.9 Maintainability Requirements

NFR-MAIN-001
The system shall use modular architecture to allow independent updates of components.

NFR-MAIN-002
Framework updates shall not require full system redeployment.

NFR-MAIN-003
Codebase shall support CI/CD pipelines.

5.10 Portability Requirements

NFR-PORT-001
The system shall operate across modern browsers.

NFR-PORT-002
The system shall be deployable in cloud-native environments.

5.11 Observability & Monitoring

NFR-OBS-001
The system shall monitor:

- API performance
- Error rates
- External integration failures
- Database health

NFR-OBS-002
Alerting mechanisms shall notify administrators of:

- System outages
- Security anomalies
- Integration failures

5.12 Business Continuity

NFR-BC-001
In case of infrastructure failure, the system shall support disaster recovery restoration.

NFR-BC-002
Critical hiring workflows shall be prioritized during partial degradation scenarios.

5.13 Ethical & AI Governance

NFR-AI-001
AI-generated recommendations shall be advisory.

NFR-AI-002
AI outputs shall not override final hiring or certification decisions.

NFR-AI-003
Bias monitoring mechanisms shall be documented for AI-assisted processes.

5.14 Non-Functional Summary

Whizard must operate as:

- A secure platform
- A scalable multi-tenant system
- A high-availability hiring infrastructure
- A compliance-aligned data ecosystem

- An audit-ready capability engine

Failure in non-functional quality directly compromises trust across:

- Students
- Colleges
- Companies
- Governance bodies

Evaluation Integrity Requirements

- Evidence must be immutable once approved.
- Evaluator identity must be stored.
- Control point validation must be auditable.
- Certification decisions must be traceable to task-level evidence.
- No automated certification without evidence.

This strengthens trust in competence validation.

6. Data Model & Entity Relationship Specification

6.1 Introduction

This section defines the conceptual and logical data model of the Whizard platform.

The data model supports:

- Multi-tenant governance
- Workforce Readiness Capability Framework (WRCF)
- Student progression tracking
- Internship & job lifecycle management
- Corporate customization
- Institutional monitoring

The data architecture shall ensure:

- Referential integrity
- Role-scoped data isolation
- Auditability
- Version control
- Historical record immutability

6.2 Core Data Domains

The Whizard data model consists of six primary domains:

1. Governance Domain
2. Capability Framework Domain
3. Student Lifecycle Domain
4. College Domain
5. Company Domain
6. Transactional & Audit Domain

6.3 Governance Domain

6.3.1 Entities

6.3.1.1 Industry

- IndustryID (PK)

- Name
- Description
- Status

6.3.1.2 ParentClub

- ParentClubID (PK)
- IndustryID (FK)
- Name
- GovernanceConfig
- Status

6.3.1.3 Club

- ClubID (PK)
- ParentClubID (FK)
- Name
- CertificationStructure
- Status

6.3.2 Relationships

- One Industry → Many ParentClubs
- One ParentClub → Many Clubs
- One Club → Many CertificationLevels

6.4 Capability Framework Domain (WRCF)

This is the intellectual backbone of Whizard.

6.4.1 Hierarchical Structure

Industry
→ Functional Group
→ Primary Work Object (PWO)
→ Secondary Work Object (SWO)
→ Capability
→ Proficiency
→ Skill
→ Task
→ Control Point

→ Evidence
→ Evaluator

6.4.2 Entities

6.4.2.1 FunctionalGroup

- FunctionalGroupID (PK)
- IndustryID (FK)
- Name
- Description

PrimaryWorkObject (PWO)

- PWOID (PK)
- FunctionalGroupID (FK)
- Name
- Description

SecondaryWorkObject (SWO)

- SWOID (PK)
- PWOID (FK)
- Name
- Description

6.4.2.3 Capability

- CapabilityID (PK)
- Name
- Description

CapabilityInstance

- CapabilityInstanceID (PK)
- FunctionalGroupID (FK)
- PWOID (FK)
- SWOID (FK)
- CapabilityID (FK)
- ProficiencyLevel
- RequiredFlag

Proficiency

- ProficiencyID (PK)
- Name
- Description

6.4.2.4 Skill

- SkillID (PK)
- CapabilityInstanceID (FK)
- Name
- Description

Task

- TaskID (PK)
- SkillID (FK)
- Description
- AssignedBy
- Deadline
- Status

TaskAssignment

- TaskAssignmentID (PK)
- TaskID (FK)
- PersonID (Student/Employee FK)
- AssignedDate
- CompletionStatus

ControlPoint

- ControlPointID (PK)
- TaskID (FK)
- ValidationCriteria
- MandatoryFlag

Evidence

- EvidenceID (PK)
- TaskAssignmentID (FK)
- UploadedBy

- EvidenceType
- FileReference
- SubmittedDate

Evaluation

- EvaluationID (PK)
- EvidenceID (FK)
- EvaluatorID
- Status
- Comments
- Timestamp

SkillCertification

- SkillCertificationID (PK)
- PersonID
- SkillID
- CertifiedBy
- CertifiedDate

CapabilityInstanceCertification

- CapabilityInstanceCertificationID (PK)
- PersonID
- CapabilityInstanceID
- Status
- CertifiedBy
- CertifiedDate

6.4.2.2 RoleDefinition

- RoleID (PK)
- FunctionalGroupID (FK)
- Name
- Description
- Version

RoleCompetenceRecord

- RoleCompetenceRecordID (PK)
- PersonID

- RoleID
- CompetenceStatus
- CompletionDate

6.4.3 Versioning Model

- RoleDefinitionVersion
- CapabilityVersion
- SkillVersion

Framework updates shall create new version records without overwriting historical data.

6.5 Certification Domain

6.5.1 Entities

6.5.1.1 CertificationLevel

- CertificationLevelID (PK)
- ClubID (FK)
- LevelNumber
- RequiredCapabilities
- MinimumThresholds

6.5.1.2 CertificationRecord

- CertificationRecordID (PK)
- StudentID (FK)
- CertificationLevelID (FK)
- AwardDate
- Status
- VersionReference

6.5.2 Relationships

- One Club → Many CertificationLevels
- One Student → Many CertificationRecords
- CertificationRecord must reference framework version used during evaluation

6.6 Student Lifecycle Domain

6.6.1 Entities

6.6.1.1 Student

- StudentID (PK)
- CollegeID (FK)
- PrimaryClubID (FK)
- AcademicYear
- Status

6.6.1.2 StudentCapabilityScore

- StudentCapabilityScoreID (PK)
- StudentID (FK)
- CapabilityID (FK)
- Score
- LastUpdated

6.6.1.3 StudentSkillScore

- StudentSkillScoreID (PK)
- StudentID (FK)
- SkillID (FK)
- Score
- EvidenceReference

6.6.1.4 Project

- ProjectID (PK)
- ClubID (FK)
- CapabilityMapping
- Status

6.6.1.5 ProjectSubmission

- SubmissionID (PK)
- StudentID (FK)
- ProjectID (FK)
- SubmissionDate
- EvaluationScore

6.6.2 Internship & Job Domain

6.6.2.1 InternshipRequisition

- InternshipID (PK)
- CompanyID (FK)
- RolePositionID (FK)
- Status

6.6.2.2 InternshipApplication

- ApplicationID (PK)
- StudentID (FK)
- InternshipID (FK)
- Status
- ApplicationDate

6.6.2.3 InternshipEvaluation

- EvaluationID (PK)
- ApplicationID (FK)
- CapabilityScoreUpdate
- FinalRating

6.6.2.4 JobRequisition

- JobID (PK)
- CompanyID (FK)
- RolePositionID (FK)
- Status

6.6.2.5 JobApplication

- JobApplicationID (PK)
- StudentID (FK)
- JobID (FK)
- Status

6.7 College Domain

6.7.1 Entities

6.7.1.1 College

- CollegeID (PK)
- Name
- Accreditation
- Status

6.7.1.2 CollegeUser

- CollegeUserID (PK)
- CollegeID (FK)
- RoleType

6.7.1.3 AcademicTranscript

- TranscriptID (PK)
- StudentID (FK)
- Semester
- GPA
- UploadedDate

6.8 Company Domain

6.8.1 Entities

6.8.1.1 Company

- CompanyID (PK)
- ParentClubID (FK)
- SubscriptionPlan
- Status

6.8.1.2 Department

- DepartmentID (PK)
- CompanyID (FK)
- FunctionalGroupMapping

6.8.1.3 RolePosition

- RolePositionID (PK)
- CompanyID (FK)

- RoleDefinitionID (FK)
- CustomThresholds

6.8.1.4 CompanyUser

- CompanyUserID (PK)
- CompanyID (FK)
- RoleType

6.8.2 Enterprise Workforce & L&D Domain

This section extends the data model to support:

- Post-hire employee lifecycle
- Internal role progression
- Skill gap analytics
- AI-personalized learning
- Simulation-based assessment
- Capability refresh tracking
- Promotion readiness scoring

6.8.3 Employee Domain

6.8.3.1 Employee

Represents a hired individual (may originate from Student entity or be externally onboarded).

Attributes:

- EmployeeID (PK)
- CompanyID (FK)
- LinkedStudentID (nullable FK)
- DepartmentID (FK)
- RolePositionID (FK)
- HireDate
- EmploymentStatus
- CareerTrack
- CreatedAt

Relationship

- One Company → Many Employees
- One Employee → One RolePosition (current)
- One Employee → Many RoleHistory records

6.8.3.2 RoleHistory

Tracks promotions and role transitions.

Attributes:

- RoleHistoryID (PK)
- EmployeeID (FK)
- PreviousRolePositionID
- NewRolePositionID
- EffectiveDate
- Reason (Promotion / Transfer / Lateral)

6.8.4 Employee Capability Domain

6.8.4.1 EmployeeCapabilityProfile

Represents capability coverage for an employee relative to role requirements.

Attributes:

- ProfileID (PK)
- EmployeeID (FK)
- RolePositionID (FK)
- CapabilityCoverageScore
- RoleReadinessScore
- GapIndex
- LastComputedDate

6.8.4.2 EmployeeCapabilityScore

Attributes:

- EmployeeCapabilityScoreID (PK)
- EmployeeID (FK)
- CapabilityID (FK)
- CurrentScore
- RequiredThreshold

- GapValue
- LastUpdated

6.8.4.3 SkillFreshnessRecord

Tracks skill decay and refresher needs.

Attributes:

- SkillFreshnessID (PK)
- EmployeeID (FK)
- SkillID (FK)
- LastValidatedDate
- FreshnessScore
- RefreshRequiredFlag

6.8.5 Skill Gap Intelligence

6.8.5.1 SkillGapReport

Generated via AI or analytics engine.

Attributes:

- ReportID (PK)
- EmployeeID (FK)
- GeneratedBy (AI / Manager)
- GeneratedDate
- GapSummary
- RiskLevel
- PromotionBlockerFlag

6.8.5.2 DepartmentCapabilityHeatmap

Aggregated analytics entity (materialized view).

Attributes:

- HeatmapID (PK)
- DepartmentID (FK)
- CapabilityID (FK)
- AverageCoverage

- GapSeverity
- LastUpdated

6.8.6 AI-Personalized Learning Domain

6.8.6.1 LearningPath

Represents a personalized capability development plan.

Attributes:

- LearningPathID (PK)
- EmployeeID (FK)
- TargetRolePositionID
- GeneratedBy (AI / TrainingManager)
- Status
- CreatedDate

6.8.6.2 LearningPathItem

Attributes:

- LearningPathItemID (PK)
- LearningPathID (FK)
- CapabilityID (FK)
- CourseID (nullable FK)
- SimulationID (nullable FK)
- PriorityLevel
- CompletionStatus

6.8.6.3 RefresherTrainingAssignment

Attributes:

- AssignmentID (PK)
- EmployeeID (FK)
- CapabilityID (FK)
- TriggerReason (SkillDecay / Regulation / RoleUpgrade)
- AssignedDate
- CompletionStatus

6.8.7 Simulation & Work-Object Validation

6.8.7.1 SimulationTemplate

Represents company-specific work-object simulation.

Attributes:

- SimulationID (PK)
- RolePositionID (FK)
- CapabilityMapping
- DifficultyLevel
- CreatedBy
- Version

6.8.7.2 SimulationAttempt

Represents employee/student execution of simulation.

Attributes:

- AttemptID (PK)
- SimulationID (FK)
- EmployeeID (nullable FK)
- StudentID (nullable FK)
- AttemptDate
- Score
- EvidenceGenerated
- EvaluationReference

6.8.7.3 SimulationEvidence

Attributes:

- EvidenceID (PK)
- AttemptID (FK)
- CapabilityID (FK)
- SkillID (FK)
- ScoreContribution
- VerifiedFlag

6.8.8 Enterprise Knowledge System

6.8.8.1 KnowledgeAsset

Represents SOPs, troubleshooting documents, training artifacts.

Attributes:

- KnowledgeAssetID (PK)
- CompanyID (FK)
- CapabilityID (FK)
- AssetType (SOP / Guide / Case / Video)
- Version
- VisibilityScope

6.8.8.2 KnowledgeQueryLog

Logs AI-powered enterprise assistant interactions.

Attributes:

- QueryLogID (PK)
- EmployeeID (FK)
- QueryText
- ResponseGenerated
- Timestamp
- LinkedKnowledgeAssetID (nullable FK)

6.8.9 Promotion Readiness Domain

6.8.9.1 PromotionEligibilityRecord

Attributes:

- PromotionRecordID (PK)
- EmployeeID (FK)
- TargetRolePositionID
- CapabilityCoverageScore
- SkillGapIndex
- SimulationPerformanceIndex
- AIRecommendationFlag

- FinalDecision
- DecisionDate

6.8.10 Extended Relationship Overview

Now the lifecycle looks like this:

Student
→ InternshipApplication
→ InternshipEvaluation
→ Employee
→ EmployeeCapabilityProfile
→ SkillGapReport
→ LearningPath
→ SimulationAttempt
→ PromotionEligibilityRecord

6.8.11 Cross-Domain Integrity Rules

1. An Employee may originate from a Student record.
2. EmployeeCapabilityProfile must reference the current RolePosition.
3. SkillGapReport must reference capability threshold logic.
4. LearningPath must map to defined CapabilityIDs.
5. SimulationAttempt must generate evidence mapped to SkillIDs.
6. PromotionEligibilityRecord must preserve a snapshot of capability state at evaluation time.
7. No framework baseline changes shall retroactively alter historical EmployeeCapabilityScore records.

6.8.12 Extended Multi-Tenant Isolation

- Employee data is isolated by CompanyID.
- Knowledge assets are company-scoped.
- Simulation templates are company-scoped.
- AI learning path generation must not expose cross-company data.
- Student-to-Employee linkage must respect consent and compliance rules.

6.8.12 Architectural Evolution Summary

With these additions, Whizard now models:

1. Pre-hire capability formation (Student domain)
2. Hiring risk reduction (Internship & Job domain)
3. Post-hire capability tracking (Employee domain)
4. Continuous L&D replacement (LearningPath & Simulation domain)
5. Promotion readiness & workforce intelligence (Promotion domain)

6.9 Communication Domain

6.9.1 Entities

6.9.1.1 Post

- PostID (PK)
- AuthorID
- ClubID (optional FK)
- VisibilityScope
- Content
- CreatedAt

6.9.1.2 Notification

- NotificationID (PK)
- RecipientID
- Type
- DeliveryChannel
- Status

6.10 Billing Domain

6.10.1 Entities

6.10.1.1 Subscription

- SubscriptionID (PK)
- CompanyID (FK)
- PlanType
- StartDate
- EndDate

6.10.1.2 Invoice

- InvoiceID (PK)
- CompanyID (FK)
- Amount
- Status
- PaymentReference

6.11 Audit & Logging Domain

6.11.1 Entities

6.11.1.1 AuditLog

- AuditID (PK)
- ActorID
- ActorType
- ActionType
- EntityAffected
- Timestamp
- PreviousValueHash
- NewValueHash

All critical state transitions must generate audit entries.

6.12 Key Relationship Summary (Cardinality Overview)

- One Industry → Many ParentClubs
- One ParentClub → Many Clubs
- One Club → Many CertificationLevels
- One Student → One PrimaryClub
- One Student → Many Certifications
- One Student → Many ProjectSubmissions
- One Student → Many InternshipApplications
- One Company → Many RolePositions
- One RolePosition → Many InternshipRequisitions
- One InternshipRequisition → Many Applications
- One Application → One Evaluation

6.13 Data Integrity Rules

1. CertificationRecord is immutable once awarded.
2. InternshipEvaluation cannot be edited after submission.
3. Framework versions must be preserved.
4. StudentSkillScore updates require evidence reference.
5. Company customization cannot modify baseline framework records.

6.14 Multi-Tenant Isolation Model

- College data is scoped by CollegeID.
- Company data is scoped by CompanyID.
- Governance data is scoped by ParentClubID.
- Student data visibility is permission-bound.
- Cross-tenant queries must be restricted.

6.15 Historical Record Preservation

The system shall:

- Preserve historical certification state.
- Preserve framework version references.
- Preserve hiring records permanently.
- Maintain audit trails for all evaluative actions.

Section 6 Summary

The Whizard Data Model is:

- Hierarchical at governance layer
- Relational at execution layer
- Version-controlled at framework layer
- Immutable at certification layer
- Multi-tenant at organizational layer
- Auditable at transactional layer

This structure ensures:

- Capability integrity
- Hiring trust

- Institutional transparency
- Industry confidence

7. End-to-End Role Competence Lifecycle

7.1 Introduction

This section defines the complete lifecycle of an individual within the Whizard ecosystem, from student development to enterprise role competence and promotion.

The lifecycle enforces the following principles:

- Capabilities belong to the Industry.
- Competence belongs to the Role.
- Tasks validate Skills.
- Evidence validates Tasks.
- Supervisors certify Skills.
- Capability Instances determine Role Readiness.
- Role Competence determines Workforce Readiness.

The lifecycle spans:

Student → Internship → Employee → Task Assignment → Evidence → Certification → Role Competence → Promotion

7.2 Phase 1 – Student Capability Formation

7.2.1 Exploration & Enrollment

1. Student selects Primary Club.
2. Student enters certification pathway aligned to Functional Group and PWO.
3. Student progresses through Capability Instances mapped to SWO + Capability + Proficiency.

7.2.2 Task-Level Learning

For each Capability Instance:

- System exposes associated Skills.
- Each Skill contains atomic Tasks.
- Tasks contain Control Points.

Student must:

- Execute Task

- Submit Evidence
- Await Evaluation

No skill is assumed. All skills must be demonstrated.

7.2.3 Skill Certification

Supervisor / Evaluator:

- Reviews evidence against Control Points
- Approves / rejects
- Certifies Skill upon approval

When all Skills under a Capability Instance are certified:

→ Capability Instance is certified.

When all required Capability Instances for a Student-level Role are certified:

→ Student is marked “Industry Role Ready (Pre-Hire)”.

7.3 Phase 2 – Internship Validation

7.3.1 Internship as Predictive Validation

During internship:

1. Student is assigned real PWO/SWO context.
2. Company Supervisor assigns operational Tasks.
3. Student submits workplace Evidence.
4. Evaluator validates performance.

Internship Evaluation updates:

- Skill scores
- Capability Instance strength
- Role readiness confidence score

Internship is treated as live environment validation of competence.

7.4 Phase 3 – Transition to Employee

7.4.1 Conversion Event

If hired:

- Student record links to Employee record.
- Certified Capability Instances transfer to EmployeeCapabilityProfile.
- RolePosition is assigned.

Employee begins enterprise lifecycle.

7.5 Phase 4 – Supervisor-Governed Enterprise Competence

7.5.1 Role Assignment

RolePosition defines:

- Required Capability Instances
- Required Proficiency Levels
- Mandatory Skills
- Control Rule set

EmployeeCapabilityProfile compares:

Required vs Achieved.

7.5.2 Task Assignment by Supervisor

Supervisor assigns:

- Tasks mapped to missing or upgrading Capability Instances
- Tasks aligned to real Primary/Secondary Work Objects

Each Task includes:

- Defined Control Points
- Deadline
- Evaluation rules

7.5.3 Evidence Submission

Employee must:

- Execute task in real operational environment

- Upload evidence (document, simulation log, approval, field verification)

Evidence is immutable once submitted.

7.5.4 Evaluation & Certification

Evaluator:

- Reviews evidence against Control Points
- Approves or rejects
- Certifies Skill upon success

System automatically updates:

- SkillCertification
- CapabilityInstanceCertification
- EmployeeCapabilityProfile

7.6 Phase 5 – Role Competence Determination

7.6.1 Capability Instance Completion Rule

Capability Instance is marked complete only when:

- All associated Skills certified
- All required proficiency levels achieved
- No mandatory Control Point pending

7.6.2 Role Competence Rule

RoleCompetenceRecord is generated when:

- All required Capability Instances for the Role are certified.

RoleCompetenceStatus = TRUE

Employee is now “Role Competent”.

7.7 Phase 6 – Continuous L&D & Skill Refresh

7.7.1 Skill Decay Monitoring

System tracks:

- SkillFreshnessRecord
- Time since last validation
- Regulatory update triggers

If freshness threshold breached:

→ RefresherTrainingAssignment is triggered.

7.7.2 AI-Generated Learning Path

AI analyzes:

- SkillGapReport
- TargetRolePosition
- Promotion eligibility

Generates:

- Personalized LearningPath
- Simulation assignments
- Refresher tasks

7.8 Phase 7 – Promotion Eligibility

7.8.1 Target Role Mapping

Promotion to higher role requires:

- New Capability Instances
- Higher Proficiency levels
- Additional SWO contexts

7.8.2 Promotion Evaluation Workflow

System generates PromotionEligibilityRecord containing:

- CapabilityCoverageScore
- SkillGapIndex
- SimulationPerformanceIndex
- Supervisor recommendations
- AI advisory signal

Final decision remains human-controlled.

7.9 Lifecycle Summary Diagram (Narrative Form)

1. Student builds Capability Instances via Tasks & Evidence.
2. Internship validates in live environment.
3. Student converts to Employee.
4. Supervisor assigns enterprise Tasks.
5. Evidence evaluated.
6. Skills certified.
7. Capability Instances completed.
8. Role Competence achieved.
9. Skill freshness monitored.
10. AI suggests upgrades.
11. Promotion evaluated via new Capability Instances.

7.10 Control & Governance Rules

1. No skill certification without evidence.
2. No capability completion without certified skills.
3. No role competence without certified capability instances.
4. No promotion without verified role competence.
5. All evaluations must be auditable.
6. AI cannot auto-certify competence.

7.11 Enterprise Impact Model

This lifecycle ensures:

- Learning spend converts into measurable competence.
- Hiring risk is reduced through verified task evidence.
- Internship becomes predictive hiring filter.
- Promotion decisions are data-backed.
- L&D becomes continuous, contextual, and AI-assisted.
- Workforce readiness is transparent and auditable.

7.12 Strategic Conclusion

Whizard replaces fragmented systems by unifying:

- Academic preparation

- Industrial validation
- Supervisor evaluation
- AI-assisted development
- Promotion governance

Into one continuous Role Competence Engine.

This lifecycle transforms workforce development from:

Course Completion → Interview Guesswork

Into:

Task Execution → Evidence Validation → Capability Certification → Role Competence → Promotion Readiness

8 – Gamification & Behavioral Design Framework

8.1 Purpose

This section defines the behavioral architecture embedded within Whizard to:

Increase long-term student engagement

Encourage sustained skill mastery

Promote healthy competition

Drive measurable employability outcomes

Reduce disengagement and drop-off rates

Align intrinsic motivation with institutional objectives

Whizard gamification shall not be entertainment-based.
It shall be outcome-driven and mastery-aligned.

8.2 Design Principles

The Gamification Framework shall be based on:

1. Progress Visibility
2. Effort Recognition
3. Excellence Amplification
4. Peer Comparison (Constructive)
5. Long-Term Identity Formation
6. Signal Value Creation

Gamification shall always reinforce:

Capability → Evidence → Certification → Employability

8.3 Core Gamification Components

8.3.1 Club Points Engine

The primary gamification currency is Club Points.

Points are derived from:

This ensures:

Mere attendance is not rewarded heavily.

Completion matters.

Demonstrated excellence carries maximum weight.

8.3.2 Employability Index

The Employability Index shall:

Update dynamically

Be visible to student

Influence ranking

Affect placement prediction signals

Visibility drives motivation.

8.4 Progression System

8.4.1 Status Tiers

Students shall be categorized into tiers:

Growth Needed

On Track

Achiever

Elite Performer (optional extension)

Thresholds configurable by central governance.

Tier visibility must be:

Visible to student

Visible to college

Visible to recruiters (where applicable)

8.4.2 Visual Progress Indicators

Student dashboard shall include:

Progress bars for certification completion

Role readiness %

Engagement percentile

Skill mastery heatmap

Placement funnel stage indicator

Progress must be:

Immediate
Clear
Quantified

8.5 Competition Architecture
8.5.1 Leaderboards

Leaderboards shall exist at multiple levels:

Club-level leaderboard

Branch-level leaderboard

College-level leaderboard

National club leaderboard

Leaderboards shall display:

Club Points

Certification count

Internship completion

Excellence recognitions

8.5.2 Challenge System

System shall run:

Weekly micro challenges

Monthly skill contests

Inter-college competitions

Industry-sponsored problem statements

Completion of challenges shall:

Award bonus Club Points

Trigger Excellence multipliers

Increase Engagement Score

8.5.3 Healthy Competition Controls

To prevent negative behaviors:

Leaderboards may show percentile rather than exact ranking in early stages.

Growth improvements rewarded, not just absolute score.

No public shaming of low performers.

8.6 Social Capital Amplification

8.6.1 Public Recognition

System shall:

Highlight top contributors

Spotlight internship success stories

Feature project demos weekly

Display “Excellence Badges”

Badges must represent:

Skill mastery

Internship excellence

Challenge winner

Mentor recognition

Peer recognition

8.6.2 Peer Endorsements

Students may endorse:

Project collaboration

Skill demonstration

Internship performance

Endorsements shall contribute partially to Excellence weight component.

8.7 Streak & Consistency Mechanics

To drive long-term engagement:

Event attendance streak

Weekly activity streak

Task submission streak

Certification progression streak

Streak break notifications shall encourage re-engagement.

8.8 Placement Funnel Nudging

Behavioral nudges shall trigger when:

Student fails to complete profile

Student misses training

Certification incomplete

Internship opportunity ignored

Nudges may include:

Push notifications

Peer comparison alerts

“Students like you completed X” messaging

Placement risk warning

8.9 Behavioral Feedback Loop

8.9.1 Feedback Model

After each activity:

Student receives:

Points gained

Skill improvement impact

Employability index delta

Ranking movement

Immediate feedback strengthens reinforcement loop.

8.9.2 Positive Reinforcement

System shall:

Celebrate milestones

Provide achievement animations

Notify club peers of achievement

Issue digital certificates instantly

8.10 Institutional Gamification

Gamification applies not only to students, but to colleges.

8.10.1 College Ranking Score

College score shall depend on:

Average Employability Index

Placement Rate

Internship Completion Rate

High Achiever %

Club Engagement Rate

Colleges compete in:

District ranking

State ranking

Zone ranking

National ranking

8.10.2 Club Health Score

Each club shall have:

Engagement Score

Certification Completion Rate

Internship Conversion Rate

Leader Activity Index

Low-performing clubs trigger governance alerts.

8.11 Anti-Gaming Safeguards

To preserve system integrity:

1. Excellence component requires supervisor validation.
2. Certification cannot be auto-awarded.
3. Internship logs require evaluator confirmation.
4. Peer endorsements capped per user.
5. Sudden abnormal point spikes trigger audit.

8.12 Long-Term Identity Design

Whizard shall encourage:

"I am a Robotics Whizard"
"I am a Data Whizard"

Students build identity around:

Club mastery

Capability strength

Recognized excellence

Identity drives sustained participation beyond rewards.

8.13 Alignment with Workforce Readiness

Gamification must always reinforce:

Task completion

Evidence submission

Capability certification

Internship validation

Promotion readiness

Points without competence shall not exist.

8.14 Behavioral Success Metrics

System shall track:

4-year retention rate

Drop-off reduction %

Certification completion %

Internship participation %

Placement funnel conversion improvement

Engagement-to-placement correlation

9 – AI & Recommendation Intelligence Framework

9.1 Purpose

This section defines the Artificial Intelligence architecture embedded within Whizard to:

- Personalize student development pathways
- Predict placement outcomes
- Identify skill gaps
- Surface high-value content
- Optimize internship matching
- Assist enterprise workforce planning
- Improve institutional performance

The AI framework shall operate as:

Advisory, Explainable, Auditable, and Human-supervised.

AI shall never auto-certify competence.

9.2 AI Architecture Overview

Whizard AI shall consist of the following modules:

1. Skill Intelligence Engine
2. Employability Prediction Engine
3. Personalized Recommendation Engine
4. Social Trend & Topic Intelligence
5. Internship & Job Matching Engine
6. Enterprise Workforce Intelligence
7. Anomaly & Risk Detection Engine

9.3 Skill Intelligence Engine

9.3.1 Capability Gap Analysis

AI shall analyze:

- CapabilityInstance completion
- SkillCertification records
- Task performance trends
- Internship evaluations

AI computes:

Gap Score per capability

Outputs:

- Weak skill clusters
- Risk indicators
- Upgrade suggestions

9.3.2 Skill Strength Profiling

AI shall generate:

- Skill Heatmaps
- Domain strength percentile
- Role readiness score

This enables targeted learning interventions.

9.4 Employability Prediction Engine

9.4.1 Placement Probability Model

AI shall predict probability of placement based on:

- Employability Index
- Internship completion
- Certification level
- Engagement score
- Branch historical placement rate
- Company demand signals

Model form:

Outputs:

- Placement likelihood %
- Risk classification
- Recommended intervention

9.4.2 Placement Risk Alerts

If placement probability < threshold:

System shall trigger:

- Skill improvement suggestion
- Training recommendation
- Internship acceleration
- Certification nudges

9.5 Personalized Recommendation Engine

9.5.1 Learning Path Recommendation

AI shall recommend:

- Courses
- Projects
- Certification attempts
- Skill challenges
- Industry events

Based on:

- Skill gaps
- Career target
- Peer success patterns
- Industry hiring trends

9.5.2 Social Feed Personalization

AI shall:

- Rank posts based on club interest
- Surface high-value internship threads
- Recommend peer collaborations
- Highlight trending challenges

Personalization shall balance:

- Relevance
- Diversity
- Opportunity exposure

9.6 Internship & Job Matching Engine

9.6.1 Capability-Based Matching

AI shall match students to internships based on:

- CapabilityInstance profile
- Skill strength
- Certification level
- Company role competence requirement

Matching Score:

Outputs:

- Ranked internship recommendations
- Fit percentage
- Skill improvement suggestions for higher match

9.6.2 Enterprise Screening Assistance

For companies: AI shall provide:

- Candidate ranking
- Capability gap report
- Internship performance prediction
- Promotion readiness signals

Human approval required before final decision.

9.7 Social Trend & Topic Intelligence

9.7.1 Trend Detection

AI shall:

- Identify trending skill discussions
- Detect rising industry topics
- Surface high-engagement posts
- Recommend event themes

9.7.2 Sentiment & Engagement Analysis

AI shall analyze:

- Student sentiment toward events
- Internship experience quality signals
- Engagement health of clubs

Low engagement triggers institutional alert.

9.8 Enterprise Workforce Intelligence

9.8.1 Department Skill Gap Heatmaps

AI shall:

- Analyze employee capability coverage
- Predict promotion readiness
- Identify workforce risk zones

9.8.2 Internal Mobility Recommendation

AI may recommend:

- Upskilling path
- Role transitions
- Lateral movement opportunities

The final decision remains human-controlled.

9.9 Institutional Intelligence

9.9.1 College Performance Forecasting

AI shall predict:

- Expected placement rate
- Internship completion trends
- Branch performance risk
- Ranking movement trajectory

9.9.2 Early Warning System

System shall detect:

- Drop in engagement
- Decline in certification completion
- Reduced internship participation

- Placement funnel bottlenecks

Alerts sent to:

- College Admin
- Club Coordinators
- Placement Cell

9.10 Explainability & Transparency

AI outputs must include:

- Reason codes
- Feature contribution summary
- Confidence score
- Historical comparison

Students must understand:

Why did they receive a recommendation?
Why are they marked at risk?

Opaque AI is prohibited.

9.11 Ethical & Bias Safeguards

AI models shall:

- Avoid demographic bias
- Exclude sensitive attributes
- Use fairness auditing tools
- Allow override by human authority
- Log all automated decisions

9.12 Continuous Learning Loop

AI models shall improve using:

- Internship outcomes
- Placement success data
- Certification completion trends
- Engagement behavior
- Employer feedback

Model retraining schedule shall be version-controlled.

9.13 AI Governance Structure

All AI components shall include:

- Version tracking
- Model performance metrics
- Drift detection
- Bias audits
- Human escalation pathway

No AI decision shall be irreversible.

Whizard is Not a reporting system
But a predictive intelligence infrastructure

It moves from Tracking what happened to Predicting what will happen and Recommending what should be done.